

# Table of Contents

|                                     |    |
|-------------------------------------|----|
| Uralstech.UAI.LiteRTLM .....        | 3  |
| ActivationDataType .....            | 5  |
| BackendNames .....                  | 6  |
| BenchmarkInfo .....                 | 8  |
| BenchmarkInfo.Turn .....            | 10 |
| ConstraintProviderType .....        | 12 |
| ConstraintType .....                | 13 |
| Conversation .....                  | 14 |
| ConversationConfig .....            | 19 |
| ConversationOptionalArgs .....      | 24 |
| DetokenizeResult .....              | 28 |
| Engine .....                        | 29 |
| EngineSettings .....                | 33 |
| InputData .....                     | 43 |
| InputDataType .....                 | 46 |
| JsonResponse .....                  | 47 |
| LiteRTLMNativeHandle .....          | 48 |
| LiteRTLMNativeLogging .....         | 50 |
| LogSeverity .....                   | 51 |
| NoRepeatNgramConfig .....           | 52 |
| RepetitionPenaltyConfig .....       | 54 |
| Responses .....                     | 57 |
| Responses.Candidate .....           | 59 |
| SamplerParams .....                 | 63 |
| SamplerType .....                   | 66 |
| Session .....                       | 67 |
| SessionConfig .....                 | 73 |
| StreamCallback .....                | 76 |
| StreamChunk .....                   | 77 |
| SuppressTokensConfig .....          | 80 |
| ThinkingConfig .....                | 82 |
| TokenUnion .....                    | 84 |
| TokenUnionType .....                | 86 |
| TokenUnions .....                   | 87 |
| TokenizeResult .....                | 88 |
| Uralstech.UAI.LiteRTLM.Native ..... | 90 |
| NativeAPI .....                     | 92 |
| NativeAPI.BenchmarkInfo .....       | 93 |

|                                          |     |
|------------------------------------------|-----|
| NativeAPI.Conversation .....             | 99  |
| NativeAPI.ConversationConfig .....       | 105 |
| NativeAPI.ConversationOptionalArgs ..... | 112 |
| NativeAPI.DetokenizeResult .....         | 118 |
| NativeAPI.Engine .....                   | 120 |
| NativeAPI.EngineSettings .....           | 124 |
| NativeAPI.InputData .....                | 137 |
| NativeAPI.JsonResponse .....             | 139 |
| NativeAPI.NoRepeatNgramConfig .....      | 141 |
| NativeAPI.RepetitionPenaltyConfig .....  | 144 |
| NativeAPI.Responses .....                | 148 |
| NativeAPI.SamplerParams .....            | 154 |
| NativeAPI.Session .....                  | 157 |
| NativeAPI.SessionConfig .....            | 165 |
| NativeAPI.StreamCallback .....           | 169 |
| NativeAPI.StreamChunk .....              | 170 |
| NativeAPI.SuppressTokensConfig .....     | 172 |
| NativeAPI.ThinkingConfig .....           | 174 |
| NativeAPI.TokenUnion .....               | 176 |
| NativeAPI.TokenUnions .....              | 179 |
| NativeAPI.TokenizeResult .....           | 181 |

# Namespace Uralstech.UAI.LiteRTLM

## Classes

[BackendNames](#)

[BenchmarkInfo](#)

[Conversation](#)

[ConversationConfig](#)

[ConversationOptionalArgs](#)

[DetokenizeResult](#)

[Engine](#)

[EngineSettings](#)

[InputData](#)

[JsonResponse](#)

[LiteRTLMNativeHandle](#)

[LiteRTLMNativeLogging](#)

[NoRepeatNgramConfig](#)

[RepetitionPenaltyConfig](#)

[Responses](#)

[Responses.Candidate](#)

[SamplerParams](#)

[Session](#)

[SessionConfig](#)

[StreamChunk](#)

[SuppressTokensConfig](#)

[ThinkingConfig](#)

[TokenUnion](#)

[TokenUnions](#)

[TokenizeResult](#)

## Structs

[BenchmarkInfo.Turn](#)

Benchmarks for a prefill or decode turn.

## Enums

[ActivationDataType](#)

[ConstraintProviderType](#)

[ConstraintType](#)

[InputDataType](#)

Represents the type of input data.

[LogSeverity](#)

Represents the log severity / level.

[SamplerType](#)

Represents the type of sampler.

[TokenUnionType](#)

Represents the type of a TokenUnion.

## Delegates

[StreamCallback](#)

Callback for streaming responses.

# Enum ActivationDataType

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public enum ActivationDataType
```

## Fields

**Float16 = 1**

Use float16 as the activation data type.

**Float32 = 0**

Use float32 as the activation data type.

**Int16 = 2**

Use int16 as the activation data type.

**Int8 = 3**

Use int8 as the activation data type.

# Class BackendNames

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public static class BackendNames
```

## Inheritance

object ← BackendNames

## Fields

### CPU

```
public const string CPU = "cpu"
```

Field Value

string

### CPUArtisan

```
public const string CPUArtisan = "cpu_artisan"
```

Field Value

string

### GPU

```
public const string GPU = "gpu"
```

Field Value

string

## GPUArtisan

```
public const string GPUArtisan = "gpu_artisan"
```

Field Value

string

## GoogleTensorArtisan

```
public const string GoogleTensorArtisan = "google_tensor_artisan"
```

Field Value

string

## NPU

```
public const string NPU = "npu"
```

Field Value

string

# Class BenchmarkInfo

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class BenchmarkInfo : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← BenchmarkInfo

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Methods

### GetDecodeTurns()

Returns the benchmark information for each decode turn.

```
public BenchmarkInfo.Turn[] GetDecodeTurns()
```

Returns

[Turn\[\]](#)

### GetPrefillTurns()

Returns the benchmark information for each prefill turn.

```
public BenchmarkInfo.Turn[] GetPrefillTurns()
```

Returns

[Turn\[\]](#)

### GetTimeToFirstToken()



Returns the time to the first token in seconds.

```
public double GetTimeToFirstToken()
```

Returns

double

The time to the first token in seconds.

Remarks

Note that the time to first token does not include initialization time. It is the sum of the prefill time for the first turn and the time spent decoding the first token.

## GetTotalInitTime()

Returns the total initialization time in seconds.

```
public double GetTotalInitTime()
```

Returns

double

The total initialization time in seconds.

## ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

# Struct BenchmarkInfo.Turn

Namespace: [Uralstech.UAI.LiteRTLM](#)

Benchmarks for a prefill or decode turn.

```
public readonly struct BenchmarkInfo.Turn
```

## Constructors

### Turn(int, double)

```
public Turn(int tokenCount, double tokensPerSecond)
```

## Parameters

**tokenCount** int

**tokensPerSecond** double

## Fields

### TokenCount

The prefill/decode token count for this turn.

```
public readonly int TokenCount
```

## Field Value

int

### TokensPerSecond

The prefill/decode tokens per second for this turn.

```
public readonly double TokensPerSecond
```

Field Value

double

# Enum ConstraintProviderType

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public enum ConstraintProviderType
```

## Fields

LlGuidance = 1

# Enum ConstraintType

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public enum ConstraintType
```

## Fields

JsonSchema = 2

None = 0

Regex = 1

# Class Conversation

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class Conversation : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← Conversation

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### Conversation(Engine, ConversationConfig?)

Creates a managed wrapper around a LiteRT LM conversation. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public Conversation(Engine engine, ConversationConfig? config = null)
```

## Parameters

**engine** [Engine](#)

The engine to create the conversation from.

**config** [ConversationConfig](#)

The conversation configuration to use. If [null](#)<sup>↗</sup>, the default configuration is used.

## Exceptions

[InvalidOperationException](#)

Thrown if the native object could not be created.

## Methods

## CancelProcess()

Cancels the ongoing inference process for asynchronous inference.

```
public void CancelProcess()
```

## Clone()

Clones a LiteRT LM conversation, duplicating its prefilled state. The caller is responsible for disposing the cloned wrapper using [Dispose\(\)](#).

```
public Conversation? Clone()
```

### Returns

[Conversation](#)

The cloned conversation wrapper, or [null](#) on failure.

## GetBenchmarkInfo()

Retrieves benchmark information from the conversation. The caller is responsible for disposing the returned wrapper.

```
public BenchmarkInfo? GetBenchmarkInfo()
```

### Returns

[BenchmarkInfo](#)

The benchmark information wrapper, or [null](#) on failure.

## GetTokenCount()

Gets the number of tokens in the conversation KV cache (prefill + decode).

```
public int GetTokenCount()
```

## Returns

int

The number of tokens, or a negative value on failure.

## ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

## RenderMessageToString(string)

Renders the message into a string according to the template.

```
public string? RenderMessageToString(string messageJson)
```

## Parameters

**messageJson** string

A JSON string representing the message to render.

## Returns

string

The rendered string, or [null](#) on failure.

## Remarks

This function does not need to be called for actual message sending, as [Send Message\(string, string?, ConversationOptionalArgs?\)](#) and [SendMessageStream\(Stream Callback, string, string?, ConversationOptionalArgs?\)](#) render messages internally.



# RenderPrefaceToString()

Renders the preface into a string according to the template.

```
public string? RenderPrefaceToString()
```

## Returns

string

The rendered string, or [null](#) on failure.

# SendMessage(string, string?, ConversationOptionalArgs?)

Sends a message to the conversation and returns the response. This is a blocking call.

```
public JsonResponse? SendMessage(string messageJson, string? extraContext = null, ConversationOptionalArgs? optionalArgs = null)
```

## Parameters

**messageJson** string

A JSON string representing the message to send.

**extraContext** string

A JSON string representing the extra context to use.

**optionalArgs** [ConversationOptionalArgs](#)

The optional arguments to use.

## Returns

[JsonResponse](#)

A JSON response wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

# SendMessageStream(StreamCallback, string, string?, ConversationOptionalArgs?)

Sends a message to the conversation and streams the response through a callback. This is a non-blocking call that invokes the callback from a background thread for each chunk.

```
public int SendMessageStream(StreamCallback callback, string messageJson, string? extraContext = null, ConversationOptionalArgs? optionalArgs = null)
```

## Parameters

**callback** [StreamCallback](#)

The callback function ([StreamCallback](#)) that receives response chunks.

**messageJson** string

A JSON string representing the message to send.

**extraContext** string

A JSON string representing the extra context to use.

**optionalArgs** [ConversationOptionalArgs](#)

The optional arguments to use.

## Returns

int

0 on success, non-zero on failure to start the stream.

# Class ConversationConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class ConversationConfig : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← ConversationConfig

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### ConversationConfig()

Creates a managed wrapper around a LiteRT LM conversation configuration. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public ConversationConfig()
```

## Exceptions

InvalidOperationException

Thrown if the native object could not be created.

## Methods

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

### SetConstraintProvider(ConstraintProviderType?)

Sets the constraint provider type for this conversation config.

```
public void SetConstraintProvider(ConstraintProviderType? providerType)
```

## Parameters

**providerType** [ConstraintProviderType?](#)

The constraint provider type to use, or [null](#) to unset.

## SetEnableConstrainedDecoding(bool)

Sets whether to enable constrained decoding for this conversation configuration.

```
public void SetEnableConstrainedDecoding(bool enableConstrainedDecoding)
```

## Parameters

**enableConstrainedDecoding** bool

Whether to enable constrained decoding.

## SetExtraContext(string)

Sets the extra context for the conversation preface.

```
public void SetExtraContext(string extraContextJson)
```

## Parameters

**extraContextJson** string

A JSON string representing the extra context object.

## SetFilterChannelContentFromKvCache(bool)

Sets whether to filter channel content from the KV cache.

```
public void SetFilterChannelContentFromKvCache(bool filterChannelContentFromKvCache)
```

## Parameters

**filterChannelContentFromKvCache** bool

Whether to filter channel content.

## SetMessages(string)

Sets the initial messages for this conversation configuration.

```
public void SetMessages(string messagesJson)
```

## Parameters

**messagesJson** string

The initial messages in JSON array format.

## SetPromptTemplate(string)

Sets the prompt template for this conversation config.

```
public void SetPromptTemplate(string promptTemplate)
```

## Parameters

**promptTemplate** string

The prompt template string (e.g. Jinja template). If not set, use the default provided by the model or the engine.

## SetSessionConfig(SessionConfig)

Sets the session config for this conversation configuration.

```
public void SetSessionConfig(SessionConfig sessionConfig)
```

## Parameters

**sessionConfig** [SessionConfig](#)

The session config to use.

## SetStreamToolCalls(bool, string)

Sets whether to stream tool call tokens.

```
public void SetStreamToolCalls(bool streamToolCalls, string channelName)
```

## Parameters

**streamToolCalls** bool

Whether to stream tool call tokens.

**channelName** string

The channel name to use for tool call tokens.

## SetSystemMessage(string)

Sets the system message for this conversation configuration.

```
public void SetSystemMessage(string systemMessageJson)
```

## Parameters

**systemMessageJson** string

The system message in JSON format.

## SetThinkingConfig(ThinkingConfig?)

Sets the thinking config for this conversation config.

```
public void SetThinkingConfig(ThinkingConfig? thinkingConfig)
```

## Parameters

**thinkingConfig** [ThinkingConfig](#)

The thinking config to set. If [null](#), clears any previously set thinking config.

## SetTools(string)

Sets the tools for this conversation configuration.

```
public void SetTools(string toolsJson)
```

## Parameters

**toolsJson** string

The tools description in JSON array format.

# Class ConversationOptionalArgs

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class ConversationOptionalArgs : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← ConversationOptionalArgs

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### ConversationOptionalArgs()

Creates a managed wrapper around the optional arguments for conversation APIs. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public ConversationOptionalArgs()
```

## Exceptions

### InvalidOperationException

Thrown if the native object could not be created.

## Methods

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

### SetConstraint(ConstraintType, string)



Sets the constraint for the conversation optional args.

```
public void SetConstraint(ConstraintType constraintType, string constraintString)
```

## Parameters

**constraintType** [ConstraintType](#)

The type of constraint.

**constraintString** string

The constraint pattern/schema/grammar string.

## SetMaxOutputTokens(int)

Sets the maximum number of output tokens for the conversation optional arguments.

```
public void SetMaxOutputTokens(int maxOutputTokens)
```

## Parameters

**maxOutputTokens** int

The maximum number of output tokens.

## SetNoRepeatNgramConfig(NoRepeatNgramConfig?)

Sets the no repeat ngram configuration for the per-turn conversation optional arguments.

```
public void SetNoRepeatNgramConfig(NoRepeatNgramConfig? noRepeatNgramConfig)
```

## Parameters

**noRepeatNgramConfig** [NoRepeatNgramConfig](#)

The no repeat ngram configuration. The contents are deep-copied when set. If [null](#), clears any previously set no repeat ngram config so no repeat ngram banning applies.

## Remarks

The configured parameters ([no\\_repeat\\_ngram\\_size](#), [window\\_size](#)) apply exclusively to the output sequence generated during the current [SendMessage\(string, string?, ConversationOptionalArgs?\)](#) or [SendMessageStream\(StreamCallback, string, string?, ConversationOptionalArgs?\)](#) call.

## SetRepetitionPenaltyConfig(RepetitionPenaltyConfig?)

Sets the repetition penalty configuration for the per-turn conversation optional arguments.

```
public void SetRepetitionPenaltyConfig(RepetitionPenaltyConfig?
    repetitionPenaltyConfig)
```

## Parameters

[repetitionPenaltyConfig](#) [RepetitionPenaltyConfig](#)

The repetition penalty configuration. The contents are deep-copied when set. If [null](#), clears any previously set repetition penalty config so no penalties apply.

## Remarks

The configured penalties ([repetition\\_penalty](#), [presence\\_penalty](#), [frequency\\_penalty](#), [window\\_size](#)) apply exclusively to the output sequence generated during the current [SendMessage\(string, string?, ConversationOptionalArgs?\)](#) or [SendMessageStream\(StreamCallback, string, string?, ConversationOptionalArgs?\)](#) call.

## SetSuppressTokensConfig(SuppressTokensConfig?)

Sets the suppress tokens configuration for the per-turn conversation optional arguments.

```
public void SetSuppressTokensConfig(SuppressTokensConfig? suppressTokensConfig)
```

## Parameters

[suppressTokensConfig](#) [SuppressTokensConfig](#)

The suppress tokens configuration. The contents are deep-copied when set. If [null](#) or if the inner token set is disabled/empty, clears any previously set suppress tokens config so no token suppression applies.

## Remarks

The configured list of suppressed tokens applies exclusively to the output sequence generated during the current [SendMessage\(string, string?, ConversationOptionalArgs?\)](#) or [SendMessageStream\(StreamCallback, string, string?, ConversationOptionalArgs?\)](#) call.

## SetThinkingConfig(ThinkingConfig?)

Sets the thinking config for the conversation optional args.

```
public void SetThinkingConfig(ThinkingConfig? thinkingConfig)
```

## Parameters

**thinkingConfig** [ThinkingConfig](#)

The thinking config to set. If [null](#), clears any previously set thinking config.

## SetVisualTokenBudget(int)

Sets the visual token budget for the conversation optional arguments.

```
public void SetVisualTokenBudget(int visualTokenBudget)
```

## Parameters

**visualTokenBudget** int

The visual token budget.

# Class DetokenizeResult

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class DetokenizeResult : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← DetokenizeResult

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Methods

### GetString()

Returns the string.

```
public string? GetString()
```

Returns

string

The detokenized string.

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

# Class Engine

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class Engine : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← Engine

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### Engine(EngineSettings)

Creates a managed wrapper around a LiteRT LM engine from the given settings. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public Engine(EngineSettings settings)
```

## Parameters

**settings** [EngineSettings](#)

The engine settings.

## Exceptions

InvalidOperationException

Thrown if the native object could not be created.

## Methods

### CreateSession(SessionConfig?)

Creates a managed wrapper around a LiteRT LM session. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public Session? CreateSession(SessionConfig? config = null)
```

## Parameters

**config** [SessionConfig](#)

The session configuration to use. If [null](#), the default session configuration is used.

## Returns

[Session](#)

The created session wrapper, or [null](#) on failure.

## Detokenize(int[])

Detokenizes token ids using the engine's tokenizer.

```
public DetokenizeResult? Detokenize(int[] tokens)
```

## Parameters

**tokens** int[]

An array of token ids to detokenize.

## Returns

[DetokenizeResult](#)

The detokenize result wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

## GetStartToken()

Returns the configured start token (BOS), if any.

```
public TokenUnion? GetStartToken()
```

## Returns

### [TokenUnion](#)

The start token wrapper, or [null](#) if none is configured. The caller is responsible for disposing the returned wrapper.

## GetStopTokens()

Returns the configured stop tokens (EOS).

```
public TokenUnions? GetStopTokens()
```

## Returns

### [TokenUnions](#)

The stop tokens wrapper, or [null](#) if none are configured. The caller is responsible for disposing the returned wrapper.

## ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

## Tokenize(string)

Tokenizes text using the engine's tokenizer.

```
public TokenizeResult? Tokenize(string text)
```

## Parameters

**text** string

The UTF-8 string to tokenize.

## Returns

### [TokenizeResult](#)

The tokenize result wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.



# Class EngineSettings

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class EngineSettings : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← EngineSettings

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### EngineSettings(int, string, string?, string?)

Creates managed engine settings from a raw file descriptor. The engine takes ownership of the file descriptor and closes it when done. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public EngineSettings(int fd, string backend, string? visionBackend = null, string? audioBackend = null)
```

## Parameters

**fd** int

The file descriptor of the model.

**backend** string

The backend to use (for example, "cpu" or "gpu").

**visionBackend** string

The vision backend to use, or [null](#) if not set.

**audioBackend** string

The audio backend to use, or [null](#) if not set.

## Remarks

See [BackendNames](#) for valid backend strings.

## Exceptions

### InvalidOperationException

Thrown if the native object could not be created.

## EngineSettings(string, string, string?, string?)

Creates managed engine settings from a model path. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public EngineSettings(string modelPath, string backend, string? visionBackend = null, string? audioBackend = null)
```

## Parameters

**modelPath** string

The path to the model file.

**backend** string

The backend to use (for example, "cpu" or "gpu").

**visionBackend** string

The vision backend to use, or [null](#) if not set.

**audioBackend** string

The audio backend to use, or [null](#) if not set.

## Remarks

See [BackendNames](#) for valid backend strings.

## Exceptions

### InvalidOperationException

Thrown if the native object could not be created.

## Methods

### EnableBenchmark()

Enables benchmarking for the engine.

```
public void EnableBenchmark()
```

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

### SetActivationDataType(ActivationDataType)

Sets the activation data type.

```
public void SetActivationDataType(ActivationDataType activationDataType)
```

## Parameters

**activationDataType** [ActivationDataType](#)

The activation data type.

### SetAudioLoraRank(int)

Sets the Audio LoRA rank for the engine.

```
public void SetAudioLoraRank(int loraRank)
```

## Parameters

**loraRank** int

The Audio LoRA rank.

## SetAudioNumThreads(int)

Sets the number of threads for the audio CPU backend.

```
public void SetAudioNumThreads(int numThreads)
```

### Parameters

**numThreads** int

The number of threads.

## SetCacheDir(string)

Sets the cache directory for the engine.

```
public void SetCacheDir(string cacheDir)
```

### Parameters

**cacheDir** string

The cache directory.

## SetEnableSpeculativeDecoding(bool)

Sets whether to enable speculative decoding.

```
public void SetEnableSpeculativeDecoding(bool enableSpeculativeDecoding)
```

### Parameters

**enableSpeculativeDecoding** bool

Whether to enable speculative decoding.

## SetEnableYNNPACK(bool)

Sets whether YNNPACK should delegate supported operations before XNNPACK.

```
public void SetEnableYNNPACK(bool enableYNNPACK)
```

### Parameters

**enableYNNPACK** bool

Whether to enable YNNPACK.

## SetGPUDecodeStepsPerSync(int)

Sets the number of decode steps per sync for the GPU backend.

```
public void SetGPUDecodeStepsPerSync(int numDecodeStepsPerSync)
```

### Parameters

**numDecodeStepsPerSync** int

The number of decode steps per sync.

### Remarks

Note: This setting is currently only supported for the Artisan GPU backend ([GPUArtisan](#)).

## SetGPUWaitForWeightUploads(bool)

Sets whether to wait for weight uploads for the GPU backend.

```
public void SetGPUWaitForWeightUploads(bool waitForWeightUploads)
```

### Parameters

`waitForWeightUploads` bool

Whether to wait for weight uploads.

## Remarks

Note: This setting is currently only supported for the Artisan GPU backend ([GPUArtisan](#)).

## SetLitertDispatchLibDir(string)

Sets the LiteRT dispatch library directory for the NPU backend.

```
public void SetLitertDispatchLibDir(string libDir)
```

## Parameters

`libDir` string

The dispatch library directory.

## SetLoraRank(int)

Sets the LoRA rank for the engine.

```
public void SetLoraRank(int loraRank)
```

## Parameters

`loraRank` int

The LoRA rank.

## SetMaxNumImages(int)

Sets the maximum number of images for the engine. This is only used for the legacy implementation of the engine.

```
public void SetMaxNumImages(int maxNumImages)
```

## Parameters

**maxNumImages** int

The maximum number of images.

## SetMaxNumTokens(int)

Sets the maximum number of tokens for the engine.

```
public void SetMaxNumTokens(int maxNumTokens)
```

## Parameters

**maxNumTokens** int

The maximum number of tokens.

## SetNumDecodeTokens(int)

Sets the number of decode tokens for benchmarking.

```
public void SetNumDecodeTokens(int numDecodeTokens)
```

## Parameters

**numDecodeTokens** int

The number of decode tokens.

## SetNumPrefillTokens(int)

Sets the number of prefill tokens for benchmarking.

```
public void SetNumPrefillTokens(int numPrefillTokens)
```

## Parameters

**numPrefillTokens** int

The number of prefill tokens.

## SetNumThreads(int)

Sets the number of threads for the CPU backend.

```
public void SetNumThreads(int numThreads)
```

## Parameters

**numThreads** int

The number of threads.

## SetParallelFileSectionLoading(bool)

Sets whether the engine should load different sections of the litertlm file in parallel. Defaults to [true](#).

```
public void SetParallelFileSectionLoading(bool parallelFileSectionLoading)
```

## Parameters

**parallelFileSectionLoading** bool

Whether to load in parallel.

## SetPrefillChunkSize(int)

Sets the prefill chunk size for the engine. Only applicable for the CPU backend with dynamic models.



```
public void SetPrefillChunkSize(int prefillChunkSize)
```

## Parameters

**prefillChunkSize** int

The prefill chunk size.

## SetSupportedAudioLoraRanks(int[])

Sets the supported Audio LoRA ranks for the engine.

```
public int SetSupportedAudioLoraRanks(int[] loraRanks)
```

## Parameters

**loraRanks** int[]

An array of supported Audio LoRA ranks.

## Returns

int

0 on success, non-zero on failure.

## SetSupportedLoraRanks(int[])

Sets the supported LoRA ranks for the engine.

```
public int SetSupportedLoraRanks(int[] loraRanks)
```

## Parameters

**loraRanks** int[]

An array of supported LoRA ranks.

## Returns

int

0 on success, non-zero on failure.

## SetUseRingbuffersLocalAttention(bool)

Sets whether to use ringbuffers for local attention KV cache.

```
public void SetUseRingbuffersLocalAttention(bool useRingbuffersLocalAttention)
```

## Parameters

`useRingbuffersLocalAttention` bool

Whether to use ringbuffers for local attention.

## Remarks

When enabled for supported models, a ringbuffer stores only necessary KV cache memory for local attention layers, minimizing memory usage. When disabled, memory is allocated for the full context length, enabling instant rewinding at the cost of higher memory usage.

Note: This feature is backend-agnostic in interface design, but currently only supported by the GPU Artisan backend. Enabling it on unsupported models or backends will be ignored with a warning.

# Class InputData

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class InputData : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← InputData

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### InputData(string)

Creates a managed wrapper around text input data. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public InputData(string data)
```

## Parameters

**data** string

The text data.

## Exceptions

### ArgumentException

Thrown if **data** is null or empty.

### InvalidOperationException

Thrown if the native object could not be created.

# InputData(InputDataType, nint, nuint)

Creates a managed wrapper around LiteRT LM input data. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public InputData(InputDataType dataType, nint nativeData, nuint size)
```

## Parameters

**dataType** [InputDataType](#)

The type of the input data.

**nativeData** nint

The data buffer. The data is copied internally.

**size** nuint

The size of the data in bytes.

## Exceptions

### ArgumentException

Thrown if **nativeData** or **size** is invalid.

### InvalidOperationException

Thrown if the native object could not be created.

# InputData(InputDataType, ReadOnlySpan<byte>)

Creates a managed wrapper around binary input data. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public InputData(InputDataType dataType, ReadOnlySpan<byte> data)
```

## Parameters

**dataType** [InputDataType](#)

The type of the input data.

`data` ReadOnlySpan<byte>

The data buffer.

## Exceptions

### ArgumentException

Thrown if `data` is empty.

### InvalidOperationException

Thrown if the native object could not be created.

## Methods

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

# Enum InputDataType

Namespace: [Uralstech.UAI.LiteRTLM](#)

Represents the type of input data.

```
public enum InputDataType
```

## Fields

Audio = 3

AudioEnd = 4

Image = 1

ImageEnd = 2

Text = 0

A UTF-8 string.

# Class JsonResponse

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class JsonResponse : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← JsonResponse

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Methods

### GetString()

Returns the JSON response string.

```
public string? GetString()
```

Returns

string

The response JSON string.

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

# Class LiteRTLMNativeHandle

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public abstract class LiteRTLMNativeHandle
```

## Inheritance

object ← LiteRTLMNativeHandle

## Derived

[BenchmarkInfo](#), [Conversation](#), [ConversationConfig](#), [ConversationOptionalArgs](#), [DetokenizeResult](#), [Engine](#), [EngineSettings](#), [InputData](#), [JsonResponse](#), [NoRepeatNgramConfig](#), [RepetitionPenaltyConfig](#), [Responses](#), [SamplerParams](#), [Session](#), [SessionConfig](#), [SuppressTokensConfig](#), [ThinkingConfig](#), [TokenUnion](#), [TokenUnions](#), [TokenizeResult](#)

## Properties

### Native

```
protected nint Native { get; init; }
```

Property Value

nint

## Methods

### Dispose()

Destroys the native LiteRT LM object.

```
public void Dispose()
```



## ~LiteRTLMLNativeHandle()

```
protected ~LiteRTLMLNativeHandle()
```

## ReleaseUnmanagedResources()

```
protected abstract void ReleaseUnmanagedResources()
```

## ThrowIfDisposed()

```
protected void ThrowIfDisposed()
```

## Operators

### implicit operator nint(LiteRTLMLNativeHandle?)

```
public static implicit operator nint(LiteRTLMLNativeHandle? handle)
```

## Parameters

**handle** [LiteRTLMLNativeHandle](#)

## Returns

nint

# Class LiteRTLMNativeLogging

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public static class LiteRTLMNativeLogging
```

## Inheritance

object ← LiteRTLMNativeLogging

## Methods

### SetMinLogLevel(LogSeverity)

Sets the minimum log level for the LiteRT LM library.

```
public static void SetMinLogLevel(LogSeverity level)
```

## Parameters

**level** [LogSeverity](#)

# Enum LogSeverity

Namespace: [Uralstech.UAI.LiteRTLM](#)

Represents the log severity / level.

```
public enum LogSeverity
```

## Fields

Debug = 1

Error = 4

Fatal = 5

Info = 2

Silent = 1000

Verbose = 0

Warning = 3

# Class NoRepeatNgramConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class NoRepeatNgramConfig : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← NoRepeatNgramConfig

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### NoRepeatNgramConfig()

Creates a LiteRT LM No Repeat Ngram Config with default values ([no\\_repeat\\_ngram\\_size](#) = 0, [window\\_size](#) = 0, which means no repeat ngram banning is disabled).

```
public NoRepeatNgramConfig()
```

## Remarks

When [no\\_repeat\\_ngram\\_size](#) is set greater than 0, any sequence of tokens (an ngram of that exact length) generated during decoding or present inside the window history can only occur at most once. If generating a candidate token would complete a repeating ngram, that candidate token's logit is set to -inf.

The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

## Exceptions

InvalidOperationException

Thrown if the native object could not be created.

## Methods

## ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

## SetNoRepeatNgramSize(int)

Sets the no repeat ngram size for the no repeat ngram config.

```
public void SetNoRepeatNgramSize(int noRepeatNgramSize)
```

### Parameters

**noRepeatNgramSize** int

The size of ngrams (consecutive token sequences) that are banned from repeating within the generation history window. If set  $> 0$ , when generating the next token would complete an already observed **no\_repeat\_ngram\_size** sequence, the logit of the candidate token is set to  $-\infty$ . If set  $\leq 0$ , no repeat ngram banning is disabled. Negative values are automatically clamped to 0 during execution.

## SetWindowSize(int)

Sets the window size for the no repeat ngram config.

```
public void SetWindowSize(int windowSize)
```

### Parameters

**windowSize** int

The maximum number of recent tokens in generation history to consider when checking for repeating ngrams. Tokens generated prior to this window are forgotten. A value of 0 means tracking all infinite generation history. Must be  $\geq 0$ ; negative values are clamped to 0. If **window\_size** is greater than 0 but less than **no\_repeat\_ngram\_size**, it is automatically clamped to **no\_repeat\_ngram\_size** so that the ngrams can fit and be tracked.

# Class RepetitionPenaltyConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class RepetitionPenaltyConfig : LiteRTLMNativeHandle
```

## Inheritance

object  $\leftarrow$  [LiteRTLMNativeHandle](#)  $\leftarrow$  RepetitionPenaltyConfig

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### RepetitionPenaltyConfig()

Creates a LiteRT LM Repetition Penalty Config with default values ([repetition\\_penalty](#) = 1.0f, [presence\\_penalty](#) = 0.0f, [frequency\\_penalty](#) = 0.0f, [window\\_size](#) = 0, which means all history with no penalties active).

```
public RepetitionPenaltyConfig()
```

## Remarks

When multiple penalties are configured and active, the order of application to output logits during decoding is:

1. Multiplicative penalty ([repetition\\_penalty](#))
2. Subtractive penalties ([presence\\_penalty](#) and [frequency\\_penalty](#)).

The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

## Exceptions

### InvalidOperationException

Thrown if the native object could not be created.

# Methods

## ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

## SetFrequencyPenalty(float)

Sets the subtractive frequency penalty for the repetition penalty config.

```
public void SetFrequencyPenalty(float frequencyPenalty)
```

### Parameters

**frequencyPenalty** float

A scalar subtracted from a token's logit, scaled linearly by the number of times that token has previously appeared inside the generated window history. Positive values discourage repetition, while negative values reward repeating tokens (OpenAI style). Defaults to 0.0f.

## SetPresencePenalty(float)

Sets the subtractive presence penalty for the repetition penalty config.

```
public void SetPresencePenalty(float presencePenalty)
```

### Parameters

**presencePenalty** float

A scalar subtracted from a token's logit if that token has appeared at least once inside the generated window history. Positive values discourage repetition, while negative values reward repeating tokens (OpenAI style). Defaults to 0.0f.

## SetRepetitionPenalty(float)

Sets the multiplicative repetition penalty for the repetition penalty config.

```
public void SetRepetitionPenalty(float repetitionPenalty)
```

## Parameters

**repetitionPenalty** float

A multiplicative penalty applied to a token's logit if that token has appeared at least once inside the generated window history (e.g., 1.0 = no penalty, 1.2 = moderate penalty). Positive logits are divided by this parameter, and negative logits are multiplied (HuggingFace style). The parameter must be  $\geq 1.0f$ ; values less than 1.0f are automatically clamped to 1.0f during execution.

## SetWindowSize(int)

Sets the window size for the repetition penalty config.

```
public void SetWindowSize(int windowSize)
```

## Parameters

**windowSize** int

The maximum number of recent tokens in generation history to consider when computing penalization. Tokens generated prior to this window are forgotten. A value of 0 means tracking all infinite generation history. Must be  $\geq 0$ ; negative values are clamped to 0 during execution.



# Class Responses

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class Responses : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← Responses

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Properties

### Count

Gets the number of elements in the collection.

```
public int Count { get; }
```

### Property Value

int

The number of elements in the collection.

### this[int]

Gets the element at the specified index in the read-only list.

```
public Responses.Candidate this[int index] { get; }
```

## Parameters

**index** int

The zero-based index of the element to get.

Property Value

[Responses.Candidate](#)

The element at the specified index in the read-only list.

## Methods

### GetEnumerator()

Returns an enumerator that iterates through the collection.

```
public IEnumerator<Responses.Candidate> GetEnumerator()
```

Returns

`IEnumerator<Responses.Candidate>`

An enumerator that can be used to iterate through the collection.

Remarks

The returned enumerator is only valid for the lifetime of this [Responses](#) object.

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

# Class Responses.Candidate

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public record Responses.Candidate
```

## Inheritance

object ← Responses.Candidate

## Constructors

Candidate(string?, bool, float, bool, int, bool, float[]?)

```
public Candidate(string? Text, bool HasScore, float Score, bool HasTokenLength, int  
TokenLength, bool HasTokenScores, float[]? TokenScores)
```

## Parameters

**Text** string

The response text.

**HasScore** bool

Does the response have a valid [Score](#)?

**Score** float

The response score.

**HasTokenLength** bool

Does the response have a valid [TokenLength](#)?

**TokenLength** int

The token length of this response.

**HasTokenScores** bool

Does the response have a valid [TokenScores](#)?

**TokenScores** float[]

The token scores for this response.

## Properties

### HasScore

Does the response have a valid [Score](#)?

```
public bool HasScore { get; init; }
```

Property Value

bool

### HasTokenLength

Does the response have a valid [TokenLength](#)?

```
public bool HasTokenLength { get; init; }
```

Property Value

bool

### HasTokenScores

Does the response have a valid [TokenScores](#)?

```
public bool HasTokenScores { get; init; }
```

Property Value

bool

## Score

The response score.

```
public float Score { get; init; }
```

### Property Value

float

## Text

The response text.

```
public string? Text { get; init; }
```

### Property Value

string

## TokenLength

The token length of this response.

```
public int TokenLength { get; init; }
```

### Property Value

int

## TokenScores

The token scores for this response.

```
public float[]? TokenScores { get; init; }
```

Property Value

float[]

# Class SamplerParams

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class SamplerParams : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← SamplerParams

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### SamplerParams(SamplerType)

Creates a managed wrapper around LiteRT LM sampler parameters for a specific sampler type. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public SamplerParams(SamplerType samplerType)
```

## Parameters

samplerType [SamplerType](#)

The sampler type to use.

## Exceptions

InvalidOperationException

Thrown if the native object could not be created.

## Methods

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

## SetSeed(int)

Sets the seed.

```
public void SetSeed(int seed)
```

### Parameters

**seed** int

## SetTemperature(float)

Sets the temperature.

```
public void SetTemperature(float temperature)
```

### Parameters

**temperature** float

## SetTopK(int)

Sets the top-k value.

```
public void SetTopK(int topK)
```

### Parameters

**topK** int

## SetTopP(float)



Sets the top-p value.

```
public void SetTopP(float topP)
```

## Parameters

**topP** float

# Enum SamplerType

Namespace: [Uralstech.UAI.LiteRTLM](#)

Represents the type of sampler.

```
public enum SamplerType
```

## Fields

Greedy = 3

Pick the token with maximum logit (i.e., argmax).

TopK = 1

Probabilistically pick among the top k tokens.

TopP = 2

Probabilistically pick among the tokens such that the sum is greater than or equal to p tokens after first performing top-k sampling.

# Class Session

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class Session : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← Session

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Methods

### CancelProcess()

Cancels the current processing in the session.

```
public void CancelProcess()
```

### GenerateContent(ReadOnlySpan<InputData>)

Generates content from the input prompt.

```
public Responses? GenerateContent(ReadOnlySpan<InputData> inputs)
```

## Parameters

**inputs** ReadOnlySpan<[InputData](#)>

An array of input wrappers representing multimodal input.

## Returns

[Responses](#)

The responses wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

## GenerateContentStream(ReadOnlySpan<InputData>, StreamCallback)

Generates content from the input prompt and streams the response through a callback. This is a non-blocking call and invokes the callback from a background thread for each chunk.

```
public int GenerateContentStream(ReadOnlySpan<InputData> inputs,
    StreamCallback callback)
```

### Parameters

**inputs** [ReadOnlySpan<InputData>](#)

An array of input wrappers representing multimodal input.

**callback** [StreamCallback](#)

The callback function ([StreamCallback](#)) that receives response chunks.

### Returns

**int**

0 on success, non-zero on failure to start the stream.

## GetBenchmarkInfo()

Retrieves benchmark information for the session. The caller is responsible for disposing the returned wrapper.

```
public BenchmarkInfo? GetBenchmarkInfo()
```

### Returns

[BenchmarkInfo](#)

The benchmark information wrapper, or [null](#) on failure.

## ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

## RewindToCheckpoint(string)

Rewinds the session to the given checkpoint label.

```
public int RewindToCheckpoint(string label)
```

### Parameters

**label** string

Label of the checkpoint to rewind to.

### Returns

int

0 on success, non-zero on failure.

## RewindToStep(int)

Rewinds the session to a specific step number.

```
public int RewindToStep(int step)
```

### Parameters

**step** int

The step number to rewind to.

### Returns

int

0 on success, non-zero on failure.

## RunDecode()

Starts decoding for the model to predict a response based on the input prompt or query added after calling [RunPrefill\(ReadOnlySpan<InputData>\)](#). This is a blocking call and returns when decoding is complete.

```
public Responses? RunDecode()
```

### Returns

#### [Responses](#)

The responses wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

## RunDecodeAsync(StreamCallback)

Starts decoding for the model to predict a response based on the input prompt or query added after calling [RunPrefill\(ReadOnlySpan<InputData>\)](#). This is a non-blocking call that streams response chunks through a callback.

```
public int RunDecodeAsync(StreamCallback callback)
```

### Parameters

**callback** [StreamCallback](#)

The callback function ([StreamCallback](#)) that receives response chunks.

### Returns

int

0 on success, non-zero on failure.

## RunPrefill(ReadOnlySpan<InputData>)

Adds the input prompt or query to the model and starts the prefill process. This is a blocking call and returns when prefill is complete.

```
public int RunPrefill(ReadOnlySpan<InputData> inputs)
```

### Parameters

**inputs** ReadOnlySpan<[InputData](#)>

An array of input wrappers representing multimodal input.

### Returns

int

0 on success, non-zero on failure.

## RunTextScoring(string[], bool)

Scores the target text after the prefill process is complete.

```
public Responses? RunTextScoring(string[] targetText, bool storeTokenLengths)
```

### Parameters

**targetText** string[]

An array of target text strings to score.

**storeTokenLengths** bool

Whether to store the token lengths of the target texts in the responses.

### Returns

[Responses](#)

The responses wrapper, or [null](#) on failure. The caller is responsible for disposing the returned wrapper.

# SaveCheckpoint(string)

Saves the current state of the session to a checkpoint with the given label.

```
public int SaveCheckpoint(string label)
```

## Parameters

**label** string

Label for the checkpoint.

## Returns

int

0 on success, non-zero on failure.



# Class SessionConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class SessionConfig : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← SessionConfig

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### SessionConfig()

Creates a managed wrapper around a LiteRT LM session configuration. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public SessionConfig()
```

## Exceptions

InvalidOperationException

Thrown if the native object could not be created.

## Methods

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

### SetApplyPromptTemplate(bool)

Sets whether to apply the prompt template for this session.

```
public void SetApplyPromptTemplate(bool applyPromptTemplate)
```

## Parameters

**applyPromptTemplate** bool

Whether to apply the prompt template.

## SetAudioLoraPath(string)

Sets the path to the audio LoRA weights file.

```
public int SetAudioLoraPath(string audioLoraPath)
```

## Parameters

**audioLoraPath** string

The path to the audio LoRA weights file.

## Returns

int

0 on success, non-zero on failure.

## SetLoraPath(string)

Sets the path to the LoRA weights file.

```
public int SetLoraPath(string loraPath)
```

## Parameters

**loraPath** string

The path to the text LoRA weights file.

## Returns

int

0 on success, non-zero on failure.

## SetMaxOutputTokens(int)

Sets the maximum number of output tokens per decode step for this session.

```
public void SetMaxOutputTokens(int maxOutputTokens)
```

## Parameters

**maxOutputTokens** int

The maximum number of output tokens.

## SetSamplerParams(SamplerParams)

Sets the sampler parameters for this session configuration.

```
public void SetSamplerParams(SamplerParams samplerParams)
```

## Parameters

**samplerParams** [SamplerParams](#)

The sampler parameters to use.

# Delegate StreamCallback

Namespace: [Uralstech.UAI.LiteRTLM](#)

Callback for streaming responses.

```
public delegate void StreamCallback(StreamChunk chunk)
```

## Parameters

**chunk** [StreamChunk](#)

The stream chunk object. It's only valid for the duration of the call.

# Class StreamChunk

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class StreamChunk
```

## Inheritance

object ← StreamChunk

## Constructors

### StreamChunk(nint)

```
public StreamChunk(nint native)
```

## Parameters

**native** nint

## Methods

### Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

### GetError()

Gets the error message associated with this chunk, if any.

```
public string? GetError()
```

Returns

string

Remarks

Returns [null](#) if there is no error.

## GetText()

Gets the text content of the chunk.

```
public string? GetText()
```

Returns

string

Returns [null](#) if there is no text content in this chunk (e.g. if it is an error or metadata-only chunk).

## IsFinal()

Returns [true](#) if this is the final chunk of the stream.

```
public bool IsFinal()
```

Returns

bool

## Operators

implicit operator nint(StreamChunk?)

```
public static implicit operator nint(StreamChunk? handle)
```

## Parameters

`handle` [StreamChunk](#)

## Returns

`nint`

# Class SuppressTokensConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class SuppressTokensConfig : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← SuppressTokensConfig

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### SuppressTokensConfig()

Creates a LiteRT LM Suppress Tokens Config with default values (an empty set of suppressed tokens, which means token suppression is disabled).

```
public SuppressTokensConfig()
```

## Remarks

When `suppress_tokens` is configured with one or more token IDs, the logits corresponding to those exact token IDs will be set directly to -inf during generation. This guarantees that those tokens can never be sampled by the model.

The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

## Exceptions

### InvalidOperationException

Thrown if the native object could not be created.

## Methods



# ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

## SetSuppressTokens(int[]?)

Sets the list of token IDs to suppress for the suppress tokens config.

```
public void SetSuppressTokens(int[]? suppressTokens)
```

### Parameters

**suppressTokens** int[]

An array of integer token IDs that should be banned from generation. During every decode step, each listed token ID's candidate logit will be forced to -inf. If **suppress\_tokens** is [null](#) or empty, any previously set suppressed tokens are cleared and token suppression is disabled.

# Class ThinkingConfig

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class ThinkingConfig : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← ThinkingConfig

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Constructors

### ThinkingConfig()

Creates a managed wrapper around a LiteRT LM thinking configuration. The caller is responsible for disposing the wrapper using [Dispose\(\)](#).

```
public ThinkingConfig()
```

## Exceptions

InvalidOperationException

Thrown if the native object could not be created.

## Methods

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

### SetEnableThinking(bool)

Sets whether thinking/reasoning generation is enabled.

```
public void SetEnableThinking(bool enableThinking)
```

## Parameters

**enableThinking** bool

Whether thinking is enabled.

## SetThinkingTokenBudget(int)

Sets the thinking token budget.

```
public void SetThinkingTokenBudget(int thinkingTokenBudget)
```

## Parameters

**thinkingTokenBudget** int

Budget for token-by-token reasoning generation (-1 for infinite).

# Class TokenUnion

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class TokenUnion : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← TokenUnion

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Methods

### GetIds(out int[]?)

Returns the token ids from a token union.

```
public int GetIds(out int[]? tokenIds)
```

## Parameters

**tokenIds** int[]

The token IDs.

## Returns

int

0 on success, non-zero if the token union does not contain token ids.

### GetString()

Returns the string value from a token union.

```
public string? GetString()
```

Returns

string

The string value, or [null](#) if the token union does not contain a string value.

## GetUnionType()

Returns the type of the token union.

```
public TokenUnionType GetUnionType()
```

Returns

[TokenUnionType](#)

The type of the token union.

## ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

# Enum TokenUnionType

Namespace: [Uralstech.UAI.LiteRTLM](#)

Represents the type of a TokenUnion.

```
public enum TokenUnionType
```

## Fields

`Ids = 1`

`String = 0`

# Class TokenUnions

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class TokenUnions : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← TokenUnions

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Methods

### GetTokenUnions()

Gets the token unions in this collection.

```
public TokenUnion[] GetTokenUnions()
```

## Returns

[TokenUnion](#)[]

The token union wrappers in the collection. The caller is responsible for disposing each returned wrapper.

### ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

# Class TokenizeResult

Namespace: [Uralstech.UAI.LiteRTLM](#)

```
public sealed class TokenizeResult : LiteRTLMNativeHandle
```

## Inheritance

object ← [LiteRTLMNativeHandle](#) ← TokenizeResult

## Inherited Members

[LiteRTLMNativeHandle.Dispose\(\)](#)

## Properties

### Length

The number of token ids from the tokenize result.

```
public int Length { get; }
```

### Property Value

int

## Methods

### AsReadOnlySpan()

Returns a span of the token ids.

```
public ReadOnlySpan<int> AsReadOnlySpan()
```

### Returns

ReadOnlySpan<int>



## Remarks

The returned span is only valid for the lifetime of this [TokenizeResult](#) object.

## CopyTo(Span<int>)

Copies the token ids into `span`.

```
public long CopyTo(Span<int> span)
```

## Parameters

`span` Span<int>

## Returns

long

The number of copied elements.

## ReleaseUnmanagedResources()

```
protected override void ReleaseUnmanagedResources()
```

# Namespace Uralstech.UAI.LiteRTLM.Native

## Classes

[NativeAPI](#)

[NativeAPI.BenchmarkInfo](#)

[NativeAPI.Conversation](#)

[NativeAPI.ConversationConfig](#)

[NativeAPI.ConversationOptionalArgs](#)

[NativeAPI.DetokenizeResult](#)

[NativeAPI.Engine](#)

[NativeAPI.EngineSettings](#)

[NativeAPI.InputData](#)

[NativeAPI.JsonResponse](#)

[NativeAPI.NoRepeatNgramConfig](#)

[NativeAPI.RepetitionPenaltyConfig](#)

[NativeAPI.Responses](#)

[NativeAPI.SamplerParams](#)

[NativeAPI.Session](#)

[NativeAPI.SessionConfig](#)

[NativeAPI.StreamChunk](#)

[NativeAPI.SuppressTokensConfig](#)

[NativeAPI.ThinkingConfig](#)

[NativeAPI.TokenUnion](#)

[NativeAPI.TokenUnions](#)

[NativeAPI.TokenizeResult](#)

# Delegates

## [NativeAPI.StreamCallback](#)

Callback for streaming responses.

# Class NativeAPI

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI
```

## Inheritance

object ← NativeAPI

## Fields

### LibLiteRTLM

```
public const string LibLiteRTLM = "litert-lm"
```

## Field Value

string

## Methods

### litert\_lm\_set\_min\_log\_level(LogSeverity)

Sets the minimum log level for the LiteRT LM library.

```
public static extern void litert_lm_set_min_log_level(LogSeverity level)
```

## Parameters

level [LogSeverity](#)

# Class NativeAPI.BenchmarkInfo

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.BenchmarkInfo
```

## Inheritance

object ← NativeAPI.BenchmarkInfo

## Methods

### litert\_lm\_benchmark\_info\_delete(nint)

Destroys a LiteRT LM Benchmark Info object.

```
public static extern void litert_lm_benchmark_info_delete(nint benchmarkInfo)
```

## Parameters

**benchmarkInfo** nint

The benchmark info to destroy.

### litert\_lm\_benchmark\_info\_get\_decode\_token\_count\_at(nint, int)

Returns the decode token count at a given turn index.

```
public static extern int litert_lm_benchmark_info_get_decode_token_count_at(nint benchmarkInfo, int index)
```

## Parameters

**benchmarkInfo** nint

The benchmark info object.

**index** int

The index of the decode turn.

Returns

int

The decode token count.

## litert\_lm\_benchmark\_info\_get\_decode\_tokens\_per\_sec\_at(nint, int)

Returns the decode tokens per second at a given turn index.

```
public static extern double  
litert_lm_benchmark_info_get_decode_tokens_per_sec_at(nint benchmarkInfo, int index)
```

Parameters

**benchmarkInfo** nint

The benchmark info object.

**index** int

The index of the decode turn.

Returns

double

The decode tokens per second.

## litert\_lm\_benchmark\_info\_get\_num\_decode\_turns(nint)

Returns the number of decode turns.

```
public static extern int litert_lm_benchmark_info_get_num_decode_turns(nint  
benchmarkInfo)
```

## Parameters

**benchmarkInfo** nint

The benchmark info object.

## Returns

int

The number of decode turns.

## litert\_lm\_benchmark\_info\_get\_num\_prefill\_turns(nint)

Returns the number of prefill turns.

```
public static extern int litert_lm_benchmark_info_get_num_prefill_turns(nint benchmarkInfo)
```

## Parameters

**benchmarkInfo** nint

The benchmark info object.

## Returns

int

The number of prefill turns.

## litert\_lm\_benchmark\_info\_get\_prefill\_token\_count\_at(nint, int)

Returns the prefill token count at a given turn index.

```
public static extern int litert_lm_benchmark_info_get_prefill_token_count_at(nint benchmarkInfo, int index)
```

## Parameters

**benchmarkInfo** nint

The benchmark info object.

**index** int

The index of the prefill turn.

## Returns

int

The prefill token count.

## litert\_lm\_benchmark\_info\_get\_prefill\_tokens\_per\_sec\_at(nint, int)

Returns the prefill tokens per second at a given turn index.

```
public static extern double  
litert_lm_benchmark_info_get_prefill_tokens_per_sec_at(nint benchmarkInfo,  
int index)
```

## Parameters

**benchmarkInfo** nint

The benchmark info object.

**index** int

The index of the prefill turn.

## Returns

double

The prefill tokens per second.



## litert\_lm\_benchmark\_info\_get\_time\_to\_first\_token(nint)

Returns the time to the first token in seconds.

```
public static extern double litert_lm_benchmark_info_get_time_to_first_token(nint benchmarkInfo)
```

### Parameters

**benchmarkInfo** nint

The benchmark info object.

### Returns

double

The time to the first token in seconds.

### Remarks

Note that the first time to token doesn't include the time for initialization. It is the sum of the prefill time for the first turn and the time spent for decoding the first token.

## litert\_lm\_benchmark\_info\_get\_total\_init\_time\_in\_second(nint)

Returns the total initialization time in seconds.

```
public static extern double  
litert_lm_benchmark_info_get_total_init_time_in_second(nint benchmarkInfo)
```

### Parameters

**benchmarkInfo** nint

The benchmark info object.

### Returns

double

The total initialization time in seconds.

# Class NativeAPI.Conversation

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.Conversation
```

## Inheritance

object ← NativeAPI.Conversation

## Methods

### litert\_lm\_conversation\_cancel\_process(nint)

Cancels the ongoing inference process, for asynchronous inference.

```
public static extern void litert_lm_conversation_cancel_process(nint conversation)
```

## Parameters

**conversation** nint

The conversation to cancel the inference for.

### litert\_lm\_conversation\_clone(nint)

Clones a LiteRT LM Conversation, duplicating its prefilled state. The caller is responsible for destroying the cloned conversation using [litert\\_lm\\_conversation\\_delete\(nint\)](#).

```
public static extern nint litert_lm_conversation_clone(nint conversation)
```

## Parameters

**conversation** nint

The conversation to clone.

## Returns

nint

A pointer to the cloned conversation, or nint.Zero on failure.

## litert\_lm\_conversation\_create(nint, nint)

Creates a LiteRT LM Conversation. The caller is responsible for destroying the conversation using [litert\\_lm\\_conversation\\_delete\(nint\)](#).

```
public static extern nint litert_lm_conversation_create(nint engine, nint config)
```

## Parameters

**engine** nint

The engine to create the conversation from.

**config** nint

The conversation config to use. If nint.Zero, the default config will be used.

## Returns

nint

A pointer to the created conversation, or nint.Zero on failure.

## litert\_lm\_conversation\_delete(nint)

Destroys a LiteRT LM Conversation.

```
public static extern void litert_lm_conversation_delete(nint conversation)
```

## Parameters

**conversation** nint

The conversation to destroy.

## litert\_lm\_conversation\_get\_benchmark\_info(nint)

Retrieves the benchmark information from the conversation. The caller is responsible for destroying the benchmark info using [litert\\_lm\\_benchmark\\_info\\_delete\(nint\)](#).

```
public static extern nint litert_lm_conversation_get_benchmark_info(nint
conversation)
```

### Parameters

**conversation** nint

The conversation to get the benchmark info from.

### Returns

nint

A pointer to the benchmark info, or nint.Zero on failure.

## litert\_lm\_conversation\_get\_token\_count(nint)

Gets the number of tokens in the conversation KV Cache (prefill + decode).

```
public static extern int litert_lm_conversation_get_token_count(nint conversation)
```

### Parameters

**conversation** nint

### Returns

int

The number of tokens, or a negative value on failure.

## litert\_lm\_conversation\_render\_message\_to\_string(nint, string)

Renders the message into a string according to the template.

```
public static extern nint litert_lm_conversation_render_message_to_string(nint
conversation, string messageJson)
```

## Parameters

**conversation** nint

The conversation instance.

**messageJson** string

A JSON string representing the message to render.

## Returns

nint

A pointer to the rendered string, or nint.Zero on failure. The returned string is owned by the Conversation object and is valid until the next call to this function or until the conversation is deleted.

## Remarks

This function does not need to be called for actual message sending, as the [litert\\_lm\\_conversation\\_send\\_message\(nint, string, string?, nint\)](#) and [litert\\_lm\\_conversation\\_send\\_message\\_stream\(nint, string, string?, nint, nint, nint\)](#) functions will handle rendering internally.

## litert\_lm\_conversation\_render\_preface\_to\_string(nint)

Renders the preface into a string according to the template.

```
public static extern nint litert_lm_conversation_render_preface_to_string(nint
conversation)
```

## Parameters

**conversation** nint

The conversation instance.

## Returns

nint

A pointer to the rendered string, or nint.Zero on failure. The returned string is owned by the Conversation object and is valid until the next call to this function or until the conversation is deleted.

## litert\_lm\_conversation\_send\_message(nint, string, string?, nint)

Sends a message to the conversation and returns the response. This is a blocking call.

```
public static extern nint litert_lm_conversation_send_message(nint conversation,
string messageJson, string? extraContext, nint optionalArgs)
```

## Parameters

**conversation** nint

The conversation to use.

**messageJson** string

A JSON string representing the message to send.

**extraContext** string

A JSON string representing the extra context to use.

**optionalArgs** nint

A pointer to the optional arguments to use.

## Returns

nint

A pointer to the JSON Response, or nint.Zero on failure. The caller is responsible for deleting the response using [litert\\_lm\\_json\\_response\\_delete\(nint\)](#).

# litert\_lm\_conversation\_send\_message\_stream(nint, string, string?, nint, nint, nint)

Sends a message to the conversation and streams the response via a callback. This is a non-blocking call that will invoke the callback from a background thread for each chunk.

```
public static extern int litert_lm_conversation_send_message_stream(nint conversation, string messageJson, string? extraContext, nint optionalArgs, nint callback, nint callbackData)
```

## Parameters

**conversation** nint

The conversation to use.

**messageJson** string

A JSON string representing the message to send.

**extraContext** string

A JSON string representing the extra context to use.

**optionalArgs** nint

A pointer to the optional arguments to use.

**callback** nint

The callback function ([NativeAPI.StreamCallback](#)) to receive response chunks.

**callbackData** nint

A pointer to user data that will be passed to the callback.

## Returns

int

0 on success, non-zero on failure to start the stream.



# Class NativeAPI.ConversationConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.ConversationConfig
```

## Inheritance

object ← NativeAPI.ConversationConfig

## Methods

### litert\_lm\_conversation\_config\_create()

Creates a LiteRT LM Conversation Config. The caller is responsible for destroying the config using [litert\\_lm\\_conversation\\_config\\_delete\(nint\)](#).

```
public static extern nint litert_lm_conversation_config_create()
```

## Returns

nint

A pointer to the created config, or nint.Zero on failure.

### litert\_lm\_conversation\_config\_delete(nint)

Destroys a LiteRT LM Conversation Config.

```
public static extern void litert_lm_conversation_config_delete(nint config)
```

## Parameters

**config** nint

The config to destroy.

## litert\_lm\_conversation\_config\_set\_constraint\_provider(nint, nint)

Sets the constraint provider type for this conversation config.

```
public static extern void litert_lm_conversation_config_set_constraint_provider(nint config, nint providerType)
```

### Parameters

**config** nint

The config to modify.

**providerType** nint

A pointer to the constraint provider type to use ([ConstraintProviderType](#)), or nint.Zero to unset.

## litert\_lm\_conversation\_config\_set\_enable\_constrained\_decoding(nint, bool)

Sets whether to enable constrained decoding for this conversation config.

```
public static extern void  
litert_lm_conversation_config_set_enable_constrained_decoding(nint config,  
bool enableConstrainedDecoding)
```

### Parameters

**config** nint

The config to modify.

**enableConstrainedDecoding** bool

Whether to enable constrained decoding.

## litert\_lm\_conversation\_config\_set\_extra\_context(nint, string)

Sets the extra context for the conversation preface.

```
public static extern void litert_lm_conversation_config_set_extra_context(nint config, string extraContextJson)
```

### Parameters

**config** nint

The config to modify.

**extraContextJson** string

A JSON string representing the extra context object.

## litert\_lm\_conversation\_config\_set\_filter\_channel\_content\_from\_kv\_cache(nint, bool)

Sets whether to filter channel content from the KV cache.

```
public static extern void litert_lm_conversation_config_set_filter_channel_content_from_kv_cache(nint config, bool filterChannelContentFromKvCache)
```

### Parameters

**config** nint

The config to modify.

**filterChannelContentFromKvCache** bool

Whether to filter channel content.

## litert\_lm\_conversation\_config\_set\_messages(nint, string)

Sets the initial messages for this conversation config.

```
public static extern void litert_lm_conversation_config_set_messages(nint config,
string messagesJson)
```

## Parameters

**config** nint

The config to modify.

**messagesJson** string

The initial messages in JSON array format.

## litert\_lm\_conversation\_config\_set\_prompt\_template(nint, string)

Sets the prompt template for this conversation config.

```
public static extern void litert_lm_conversation_config_set_prompt_template(nint
config, string promptTemplate)
```

## Parameters

**config** nint

The config to modify.

**promptTemplate** string

The prompt template string (e.g. Jinja template). If not set, use the default provided by the model or the engine.

## litert\_lm\_conversation\_config\_set\_session\_config(nint, nint)

Sets the session config for this conversation config.

```
public static extern void litert_lm_conversation_config_set_session_config(nint
config, nint sessionConfig)
```

## Parameters

**config** nint

The config to modify.

**sessionConfig** nint

The session config to use.

## litert\_lm\_conversation\_config\_set\_stream\_tool\_calls(nint, bool, string)

Sets whether to stream tool call tokens.

```
public static extern void litert_lm_conversation_config_set_stream_tool_calls(nint
config, bool streamToolCalls, string channelName)
```

## Parameters

**config** nint

The config to modify.

**streamToolCalls** bool

Whether to stream tool call tokens.

**channelName** string

The channel name to use for tool call tokens.

## litert\_lm\_conversation\_config\_set\_system\_message(nint, string)

Sets the system message for this conversation config.

```
public static extern void litert_lm_conversation_config_set_system_message(nint
config, string systemMessageJson)
```

## Parameters

**config** nint

The config to modify.

**systemMessageJson** string

The system message in JSON format.

## litert\_lm\_conversation\_config\_set\_thinking\_config(nint, nint)

Sets the thinking config for this conversation config.

```
public static extern void litert_lm_conversation_config_set_thinking_config(nint
config, nint thinkingConfig)
```

## Parameters

**config** nint

The config to modify.

**thinkingConfig** nint

The thinking config to set. If nint.Zero, clears any previously set thinking config.

## litert\_lm\_conversation\_config\_set\_tools(nint, string)

Sets the tools for this conversation config.

```
public static extern void litert_lm_conversation_config_set_tools(nint config,
string toolsJson)
```

## Parameters

`config` nint

The config to modify.

`toolsJson` string

The tools description in JSON array format.

# Class NativeAPI.ConversationOptionalArgs

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.ConversationOptionalArgs
```

## Inheritance

object ← NativeAPI.ConversationOptionalArgs

## Methods

### litert\_lm\_conversation\_optional\_args\_create()

Creates a LiteRT LM Conversation Optional Args. The caller is responsible for destroying the optional args using [litert\\_lm\\_conversation\\_optional\\_args\\_delete\(nint\)](#).

```
public static extern nint litert_lm_conversation_optional_args_create()
```

## Returns

nint

A pointer to the created optional args, or nint.Zero on failure.

### litert\_lm\_conversation\_optional\_args\_delete(nint)

Destroys a LiteRT LM Conversation Optional Args.

```
public static extern void litert_lm_conversation_optional_args_delete(nint optionalArgs)
```

## Parameters

**optionalArgs** nint

The optional args to destroy.



## litert\_lm\_conversation\_optional\_args\_set\_constraint(nint, ConstraintType, string)

Sets the constraint for the conversation optional args.

```
public static extern void litert_lm_conversation_optional_args_set_constraint(nint optionalArgs, ConstraintType constraintType, string constraintString)
```

### Parameters

**optionalArgs** nint

The optional args to modify.

**constraintType** [ConstraintType](#)

The type of constraint.

**constraintString** string

The constraint pattern/schema/grammar string.

## litert\_lm\_conversation\_optional\_args\_set\_max\_output\_tokens(nint, int)

Sets the maximum number of output tokens for the conversation optional args.

```
public static extern void  
litert_lm_conversation_optional_args_set_max_output_tokens(nint optionalArgs,  
int maxOutputTokens)
```

### Parameters

**optionalArgs** nint

The optional args to modify.

**maxOutputTokens** int

The maximum number of output tokens.

## litert\_lm\_conversation\_optional\_args\_set\_no\_repeat\_ngram\_config(nint, nint)

Sets the no repeat ngram configuration for the per-turn conversation optional arguments ([OptionalArgs](#)).

```
public static extern void
litert_lm_conversation_optional_args_set_no_repeat_ngram_config(nint optionalArgs,
nint noRepeatNgramConfig)
```

### Parameters

[optionalArgs](#) nint

The optional arguments structure ([OptionalArgs](#)) to modify.

[noRepeatNgramConfig](#) nint

The no repeat ngram configuration struct ([LiteRtLmNoRepeatNgramConfig](#)) created via [litert\\_lm\\_no\\_repeat\\_ngram\\_config\\_create\(\)](#). The contents are deep-copied when set. If nint. Zero, clears any previously set no repeat ngram config so no repeat ngram banning applies.

### Remarks

The configured parameters ([no\\_repeat\\_ngram\\_size](#), [window\\_size](#)) apply exclusively to the output sequence generated during the current [send\\_message](#) or [send\\_message\\_async](#) call.

## litert\_lm\_conversation\_optional\_args\_set\_repetition\_penalty\_config(nint, nint)

Sets the repetition penalty configuration for the per-turn conversation optional arguments ([OptionalArgs](#)).

```
public static extern void
litert_lm_conversation_optional_args_set_repetition_penalty_config(nint
optionalArgs, nint repetitionPenaltyConfig)
```

### Parameters

`optionalArgs` `nint`

The optional arguments structure (`OptionalArgs`) to modify.

`repetitionPenaltyConfig` `nint`

The repetition penalty configuration struct (`LiteRtLmRepetitionPenaltyConfig`) created via [litert\\_lm\\_repetition\\_penalty\\_config\\_create\(\)](#). The contents are deep-copied when set. If `nint.Zero`, clears any previously set repetition penalty config so no penalties apply.

## Remarks

The configured penalties (`repetition_penalty`, `presence_penalty`, `frequency_penalty`, `window_size`) apply exclusively to the output sequence generated during the current `send_message` or `send_message_async` call.

## litert\_lm\_conversation\_optional\_args\_set\_suppress\_tokens\_config(nint, nint)

Sets the suppress tokens configuration for the per-turn conversation optional arguments (`OptionalArgs`).

```
public static extern void
litert_lm_conversation_optional_args_set_suppress_tokens_config(nint optionalArgs,
nint suppressTokensConfig)
```

## Parameters

`optionalArgs` `nint`

The optional arguments structure (`OptionalArgs`) to modify.

`suppressTokensConfig` `nint`

The suppress tokens configuration struct (`LiteRtLmSuppressTokensConfig`) created via [litert\\_lm\\_suppress\\_tokens\\_config\\_create\(\)](#). The contents are deep-copied when set. If `nint.Zero` or if the inner token set is disabled/empty, clears any previously set suppress tokens config so no token suppression applies.

## Remarks

The configured list of suppressed tokens applies exclusively to the output sequence generated during the current `send_message` or `send_message_async` call.

## litert\_lm\_conversation\_optional\_args\_set\_thinking\_config(nint, nint)

Sets the thinking config for the conversation optional args.

```
public static extern void
litert_lm_conversation_optional_args_set_thinking_config(nint optionalArgs,
nint thinkingConfig)
```

### Parameters

`optionalArgs` nint

The optional args to modify.

`thinkingConfig` nint

The thinking config to set. If `nint.Zero`, clears any previously set thinking config.

## litert\_lm\_conversation\_optional\_args\_set\_visual\_token\_budget(nint, int)

Sets the visual token budget for the conversation optional args.

```
public static extern void
litert_lm_conversation_optional_args_set_visual_token_budget(nint optionalArgs,
int visualTokenBudget)
```

### Parameters

`optionalArgs` nint

The optional args to modify.

`visualTokenBudget` int

The visual token budget.

# Class NativeAPI.DetokenizeResult

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.DetokenizeResult
```

## Inheritance

object ← NativeAPI.DetokenizeResult

## Methods

### litert\_lm\_detokenize\_result\_delete(nint)

Destroys a LiteRT LM Detokenize Result.

```
public static extern void litert_lm_detokenize_result_delete(nint result)
```

## Parameters

**result** nint

The detokenize result to destroy.

### litert\_lm\_detokenize\_result\_get\_string(nint)

Returns the string from a detokenize result.

```
public static extern nint litert_lm_detokenize_result_get_string(nint result)
```

## Parameters

**result** nint

The detokenize result.

## Returns

nint

The detokenized string. The returned string is owned by the DetokenizeResult object and is valid only for its lifetime.

# Class NativeAPI.Engine

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.Engine
```

## Inheritance

object ← NativeAPI.Engine

## Methods

### litert\_lm\_engine\_create(nint)

Creates a LiteRT LM Engine from the given settings. The caller is responsible for destroying the engine using [litert\\_lm\\_engine\\_delete\(nint\)](#).

```
public static extern nint litert_lm_engine_create(nint settings)
```

## Parameters

**settings** nint

The engine settings.

## Returns

nint

A pointer to the created engine, or nint.Zero on failure.

### litert\_lm\_engine\_create\_session(nint, nint)

Creates a LiteRT LM Session. The caller is responsible for destroying the session using [litert\\_lm\\_session\\_delete\(nint\)](#).

```
public static extern nint litert_lm_engine_create_session(nint engine, nint config)
```



## Parameters

**engine** nint

The engine to create the session from.

**config** nint

The session config of the session. If nint.Zero, use the default session config.

## Returns

nint

A pointer to the created session, or nint.Zero on failure.

## litert\_lm\_engine\_delete(nint)

Destroys a LiteRT LM Engine.

```
public static extern void litert_lm_engine_delete(nint engine)
```

## Parameters

**engine** nint

The engine to destroy.

## litert\_lm\_engine\_detokenize(nint, int[], nuint)

Detokenizes token ids using the engine's tokenizer.

```
public static extern nint litert_lm_engine_detokenize(nint engine, int[] tokens, nuint numTokens)
```

## Parameters

**engine** nint

The engine instance.

**tokens** int[]

An array of token ids to detokenize.

**numTokens** nuint

The number of token ids in the array.

## Returns

nint

A pointer to the detokenize result, or nint.Zero on failure. The caller is responsible for deleting the result using [litert\\_lm\\_detokenize\\_result\\_delete\(nint\)](#).

## litert\_lm\_engine\_get\_start\_token(nint)

Returns the configured start token (BOS), if any.

```
public static extern nint litert_lm_engine_get_start_token(nint engine)
```

## Parameters

**engine** nint

The engine instance.

## Returns

nint

A pointer to the start token, or nint.Zero if none configured. The caller is responsible for deleting the result using [litert\\_lm\\_token\\_union\\_delete\(nint\)](#).

## litert\_lm\_engine\_get\_stop\_tokens(nint)

Returns the configured stop tokens (EOS).

```
public static extern nint litert_lm_engine_get_stop_tokens(nint engine)
```

## Parameters

**engine** nint

The engine instance.

## Returns

nint

A pointer to the stop tokens collection, or nint.Zero if none configured. The caller is responsible for deleting the result using [litert\\_lm\\_token\\_unions\\_delete\(nint\)](#).

## litert\_lm\_engine\_tokenize(nint, string)

Tokenizes text using the engine's tokenizer.

```
public static extern nint litert_lm_engine_tokenize(nint engine, string text)
```

## Parameters

**engine** nint

The engine instance.

**text** string

The UTF-8 string to tokenize.

## Returns

nint

A pointer to the tokenize result, or nint.Zero on failure. The caller is responsible for deleting the result using [litert\\_lm\\_tokenize\\_result\\_delete\(nint\)](#).

# Class NativeAPI.EngineSettings

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.EngineSettings
```

## Inheritance

object ← NativeAPI.EngineSettings

## Methods

litert\_lm\_engine\_settings\_create(string, string, string?, string?)

Creates LiteRT LM Engine Settings. The caller is responsible for destroying the settings using [litert\\_lm\\_engine\\_settings\\_delete\(nint\)](#).

```
public static extern nint litert_lm_engine_settings_create(string modelPath, string backendStr, string? visionBackendStr, string? audioBackendStr)
```

## Parameters

**modelPath** string

The path to the model file.

**backendStr** string

The backend to use (e.g., "cpu", "gpu").

**visionBackendStr** string

The vision backend to use, or [null](#) if not set.

**audioBackendStr** string

The audio backend to use, or [null](#) if not set.

## Returns

nint

A pointer to the created settings, or nint.Zero on failure.

## litert\_lm\_engine\_settings\_create\_from\_raw\_file\_descriptor(int, string, string?, string?)

Creates LiteRT LM Engine Settings from a raw file descriptor. The engine takes ownership of the file descriptor and will close it when done. The caller is responsible for destroying the settings using [litert\\_lm\\_engine\\_settings\\_delete\(nint\)](#).

```
public static extern nint  
litert_lm_engine_settings_create_from_raw_file_descriptor(int fd, string backendStr,  
string? visionBackendStr, string? audioBackendStr)
```

### Parameters

**fd** int

The file descriptor of the model.

**backendStr** string

The backend to use (e.g., "cpu", "gpu").

**visionBackendStr** string

The vision backend to use, or [null](#) if not set.

**audioBackendStr** string

The audio backend to use, or [null](#) if not set.

### Returns

nint

A pointer to the created settings, or nint.Zero on failure.

## litert\_lm\_engine\_settings\_delete(nint)

Destroys LiteRT LM Engine Settings.

```
public static extern void litert_lm_engine_settings_delete(nint settings)
```

## Parameters

**settings** nint

The settings to destroy.

## litert\_lm\_engine\_settings\_enable\_benchmark(nint)

Enables benchmarking for the engine.

```
public static extern void litert_lm_engine_settings_enable_benchmark(nint settings)
```

## Parameters

**settings** nint

The engine settings.

## litert\_lm\_engine\_settings\_set\_activation\_data\_type(nint, ActivationDataType)

Sets the activation data type.

```
public static extern void litert_lm_engine_settings_set_activation_data_type(nint settings, ActivationDataType activationDataType)
```

## Parameters

**settings** nint

The engine settings.

**activationDataType** [ActivationDataType](#)

The activation data type.

## litert\_lm\_engine\_settings\_set\_audio\_lora\_rank(nint, int)

Sets the Audio LoRA rank for the engine.

```
public static extern void litert_lm_engine_settings_set_audio_lora_rank(nint
settings, int loraRank)
```

### Parameters

**settings** nint

The engine settings.

**loraRank** int

The Audio LoRA rank.

## litert\_lm\_engine\_settings\_set\_audio\_num\_threads(nint, int)

Sets the number of threads for the audio CPU backend.

```
public static extern void litert_lm_engine_settings_set_audio_num_threads(nint
settings, int numThreads)
```

### Parameters

**settings** nint

The engine settings.

**numThreads** int

The number of threads.

## litert\_lm\_engine\_settings\_set\_cache\_dir(nint, string)

Sets the cache directory for the engine.

```
public static extern void litert_lm_engine_settings_set_cache_dir(nint settings,
string cacheDir)
```

## Parameters

**settings** nint

The engine settings.

**cacheDir** string

The cache directory.

## litert\_lm\_engine\_settings\_set\_enable\_speculative\_decoding(nint, bool)

Sets whether to enable speculative decoding.

```
public static extern void
litert_lm_engine_settings_set_enable_speculative_decoding(nint settings,
bool enableSpeculativeDecoding)
```

## Parameters

**settings** nint

The engine settings.

**enableSpeculativeDecoding** bool

Whether to enable speculative decoding.

## litert\_lm\_engine\_settings\_set\_enable\_ynnpack(nint, bool)

Sets whether YNNPACK should delegate supported operations before XNNPACK.



```
public static extern void litert_lm_engine_settings_set_enable_ynnpack(nint
settings, bool enableYnnpack)
```

## Parameters

**settings** nint

The engine settings.

**enableYnnpack** bool

Whether to enable YNNPACK.

## litert\_lm\_engine\_settings\_set\_gpu\_decode\_steps\_per\_sync(nint, int)

Sets the number of decode steps per sync for the GPU backend.

```
public static extern void
litert_lm_engine_settings_set_gpu_decode_steps_per_sync(nint settings,
int numDecodeStepsPerSync)
```

## Parameters

**settings** nint

The engine settings.

**numDecodeStepsPerSync** int

The number of decode steps per sync.

## Remarks

Note: This setting is currently only supported for the Artisan GPU backend ([GPUArtisan](#)).

## litert\_lm\_engine\_settings\_set\_gpu\_wait\_for\_weight\_uploads(nint, bool)

Sets whether to wait for weight uploads for the GPU backend.

```
public static extern void  
litert_lm_engine_settings_set_gpu_wait_for_weight_uploads(nint settings,  
bool waitForWeightUploads)
```

## Parameters

**settings** nint

The engine settings.

**waitForWeightUploads** bool

Whether to wait for weight uploads.

## Remarks

Note: This setting is currently only supported for the Artisan GPU backend ([GPUArtisan](#)).

## litert\_lm\_engine\_settings\_set\_litert\_dispatch\_lib\_dir(nint, string)

Sets the LiteRT dispatch library directory for NPU backend.

```
public static extern void litert_lm_engine_settings_set_litert_dispatch_lib_dir(nint  
settings, string libDir)
```

## Parameters

**settings** nint

The engine settings.

**libDir** string

The dispatch library directory.

## litert\_lm\_engine\_settings\_set\_lora\_rank(nint, int)

Sets the LoRA rank for the engine.

```
public static extern void litert_lm_engine_settings_set_lora_rank(nint settings,  
int loraRank)
```

## Parameters

**settings** nint

The engine settings.

**loraRank** int

The LoRA rank.

## litert\_lm\_engine\_settings\_set\_max\_num\_images(nint, int)

Sets the maximum number of images for the engine. This is only used for the legacy implementation of the engine.

```
public static extern void litert_lm_engine_settings_set_max_num_images(nint  
settings, int maxNumImages)
```

## Parameters

**settings** nint

The engine settings.

**maxNumImages** int

The maximum number of images.

## litert\_lm\_engine\_settings\_set\_max\_num\_tokens(nint, int)

Sets the maximum number of tokens for the engine.

```
public static extern void litert_lm_engine_settings_set_max_num_tokens(nint
settings, int maxNumTokens)
```

## Parameters

**settings** nint

The engine settings.

**maxNumTokens** int

The maximum number of tokens.

## litert\_lm\_engine\_settings\_set\_num\_decode\_tokens(nint, int)

Sets the number of decode tokens for benchmarking.

```
public static extern void litert_lm_engine_settings_set_num_decode_tokens(nint
settings, int numDecodeTokens)
```

## Parameters

**settings** nint

The engine settings.

**numDecodeTokens** int

The number of decode tokens.

## litert\_lm\_engine\_settings\_set\_num\_prefill\_tokens(nint, int)

Sets the number of prefill tokens for benchmarking.

```
public static extern void litert_lm_engine_settings_set_num_prefill_tokens(nint
settings, int numPrefillTokens)
```

## Parameters

**settings** nint

The engine settings.

**numPrefillTokens** int

The number of prefill tokens.

## litert\_lm\_engine\_settings\_set\_num\_threads(nint, int)

Sets the number of threads for the CPU backend.

```
public static extern void litert_lm_engine_settings_set_num_threads(nint settings,
int numThreads)
```

## Parameters

**settings** nint

The engine settings.

**numThreads** int

The number of threads.

## litert\_lm\_engine\_settings\_set\_parallel\_file\_section\_loading(nint, bool)

Sets whether the engine should load different sections of the litertlm file in parallel. Defaults to [true](#).

```
public static extern void
litert_lm_engine_settings_set_parallel_file_section_loading(nint settings,
bool parallelFileSectionLoading)
```

## Parameters

**settings** nint

The engine settings.

`parallelFileSectionLoading` bool

Whether to load in parallel.

## `litert_lm_engine_settings_set_prefill_chunk_size(nint, int)`

Sets the prefill chunk size for the engine. Only applicable for CPU backend with dynamic models.

```
public static extern void litert_lm_engine_settings_set_prefill_chunk_size(nint settings, int prefillChunkSize)
```

### Parameters

`settings` nint

The engine settings.

`prefillChunkSize` int

The prefill chunk size.

## `litert_lm_engine_settings_set_supported_audio_lora_ranks(nint, int[], nuint)`

Sets the supported Audio LoRA ranks for the engine.

```
public static extern int  
litert_lm_engine_settings_set_supported_audio_lora_ranks(nint settings, int[]  
loraRanks, nuint numRanks)
```

### Parameters

`settings` nint

The engine settings.

**loraRanks** int[]

An array of supported Audio LoRA ranks.

**numRanks** nuint

The number of ranks in the array.

## Returns

int

0 on success, non-zero on failure.

## litert\_lm\_engine\_settings\_set\_supported\_lora\_ranks(nint, int[], nuint)

Sets the supported LoRA ranks for the engine.

```
public static extern int litert_lm_engine_settings_set_supported_lora_ranks(nint settings, int[] loraRanks, nuint numRanks)
```

## Parameters

**settings** nint

The engine settings.

**loraRanks** int[]

An array of supported LoRA ranks.

**numRanks** nuint

The number of ranks in the array.

## Returns

int

0 on success, non-zero on failure.

# litert\_lm\_engine\_settings\_set\_use\_ringbuffers\_local\_attention(nint, bool)

Sets whether to use ringbuffers for local attention KV cache.

```
public static extern void  
litert_lm_engine_settings_set_use_ringbuffers_local_attention(nint settings, bool  
useRingbuffersLocalAttention)
```

## Parameters

**settings** nint

The engine settings.

**useRingbuffersLocalAttention** bool

Whether to use ringbuffers for local attention.

## Remarks

When enabled for supported models, a ringbuffer stores only necessary KV cache memory for local attention layers, minimizing memory usage. When disabled, memory is allocated for the full context length, enabling instant rewinding at the cost of higher memory usage.

Note: This feature is backend-agnostic in interface design, but currently only supported by the GPU Artisan backend. Enabling it on unsupported models or backends will be ignored with a warning.



# Class NativeAPI.InputData

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.InputData
```

## Inheritance

object ← NativeAPI.InputData

## Methods

### litert\_lm\_input\_data\_create(InputDataType, nint, nuint)

Creates a LiteRT LM Input Data. The caller is responsible for destroying the input data using [litert\\_lm\\_input\\_data\\_delete\(nint\)](#).

```
public static extern nint litert_lm_input_data_create(InputDataType type, nint data,
nuint size)
```

## Parameters

**type** [InputDataType](#)

The type of the input data.

**data** nint

The data pointer. The data is copied internally.

**size** nuint

The size of the data in bytes.

## Returns

nint

A pointer to the created input data, or nint.Zero on failure.

# litert\_lm\_input\_data\_delete(nint)

Destroys a LiteRT LM Input Data.

```
public static extern void litert_lm_input_data_delete(nint inputData)
```

## Parameters

**inputData** nint

The input data to destroy.

# Class NativeAPI.JsonResponse

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.JsonResponse
```

## Inheritance

object ← NativeAPI.JsonResponse

## Methods

### litert\_lm\_json\_response\_delete(nint)

Destroys a LiteRT LM Json Response object.

```
public static extern void litert_lm_json_response_delete(nint response)
```

## Parameters

**response** nint

The response to destroy.

### litert\_lm\_json\_response\_get\_string(nint)

Returns the JSON response string from a response object.

```
public static extern nint litert_lm_json_response_get_string(nint response)
```

## Parameters

**response** nint

The response object.

## Returns

nint

The response JSON string. The returned string is owned by the JsonResponse object and is valid only for its lifetime. Returns nint.Zero if **response** is nint.Zero.

# Class NativeAPI.NoRepeatNgramConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.NoRepeatNgramConfig
```

## Inheritance

object ← NativeAPI.NoRepeatNgramConfig

## Methods

### litert\_lm\_no\_repeat\_ngram\_config\_create()

Creates a LiteRT LM No Repeat Ngram Config with default values (`no_repeat_ngram_size` = 0, `window_size` = 0, which means no repeat ngram banning is disabled).

```
public static extern nint litert_lm_no_repeat_ngram_config_create()
```

## Returns

nint

A pointer to the created config, or nint.Zero on failure.

## Remarks

When `no_repeat_ngram_size` is set greater than 0, any sequence of tokens (an ngram of that exact length) generated during decoding or present inside the window history can only occur at most once. If generating a candidate token would complete a repeating ngram, that candidate token's logit is set to -inf.

The caller is responsible for destroying the config using [litert\\_lm\\_no\\_repeat\\_ngram\\_config\\_delete\(nint\)](#).

### litert\_lm\_no\_repeat\_ngram\_config\_delete(nint)

Destroys a LiteRT LM No Repeat Ngram Config.

```
public static extern void litert_lm_no_repeat_ngram_config_delete(nint config)
```

## Parameters

**config** nint

The config to destroy.

## litert\_lm\_no\_repeat\_ngram\_config\_set\_no\_repeat\_ngram\_size(nint, int)

Sets the no repeat ngram size for the no repeat ngram config.

```
public static extern void  
litert_lm_no_repeat_ngram_config_set_no_repeat_ngram_size(nint config,  
int noRepeatNgramSize)
```

## Parameters

**config** nint

The config to modify.

**noRepeatNgramSize** int

The size of ngrams (consecutive token sequences) that are banned from repeating within the generation history window. If set > 0, when generating the next token would complete an already observed **no\_repeat\_ngram\_size** sequence, the logit of the candidate token is set to -inf. If set <= 0, no repeat ngram banning is disabled. Negative values are automatically clamped to 0 during execution.

## litert\_lm\_no\_repeat\_ngram\_config\_set\_window\_size(nint, int)

Sets the window size for the no repeat ngram config.

```
public static extern void litert_lm_no_repeat_ngram_config_set_window_size(nint  
config, int windowSize)
```

## Parameters

`config` nint

The config to modify.

`windowSize` int

The maximum number of recent tokens in generation history to consider when checking for repeating ngrams. Tokens generated prior to this window are forgotten. A value of 0 means tracking all infinite generation history. Must be  $\geq 0$ ; negative values are clamped to 0. If `window_size` is greater than 0 but less than `no_repeat_ngram_size`, it is automatically clamped to `no_repeat_ngram_size` so that the ngrams can fit and be tracked.

# Class NativeAPI.RepetitionPenaltyConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.RepetitionPenaltyConfig
```

## Inheritance

object ← NativeAPI.RepetitionPenaltyConfig

## Methods

### litert\_lm\_repetition\_penalty\_config\_create()

Creates a LiteRT LM Repetition Penalty Config with default values (`repetition_penalty` = 1.0f, `presence_penalty` = 0.0f, `frequency_penalty` = 0.0f, `window_size` = 0, which means all history with no penalties active).

```
public static extern nint litert_lm_repetition_penalty_config_create()
```

## Returns

nint

A pointer to the created config, or nint.Zero on failure.

## Remarks

When multiple penalties are configured and active, the order of application to output logits during decoding is:

1. Multiplicative penalty (`repetition_penalty`)
2. Subtractive penalties (`presence_penalty` and `frequency_penalty`).

The caller is responsible for destroying the config using [litert\\_lm\\_repetition\\_penalty\\_config\\_delete\(nint\)](#).

### litert\_lm\_repetition\_penalty\_config\_delete(nint)



Destroys a LiteRT LM Repetition Penalty Config.

```
public static extern void litert_lm_repetition_penalty_config_delete(nint config)
```

## Parameters

**config** nint

The config to destroy.

## litert\_lm\_repetition\_penalty\_config\_set\_frequency\_penalty(nint, float)

Sets the subtractive frequency penalty for the repetition penalty config.

```
public static extern void  
litert_lm_repetition_penalty_config_set_frequency_penalty(nint config,  
float frequencyPenalty)
```

## Parameters

**config** nint

The config to modify.

**frequencyPenalty** float

A scalar subtracted from a token's logit, scaled linearly by the number of times that token has previously appeared inside the generated window history. Positive values discourage repetition, while negative values reward repeating tokens (OpenAI style). Defaults to 0.0f.

## litert\_lm\_repetition\_penalty\_config\_set\_presence\_penalty(nint, float)

Sets the subtractive presence penalty for the repetition penalty config.

```
public static extern void  
litert_lm_repetition_penalty_config_set_presence_penalty(nint config,
```

```
float presencePenalty)
```

## Parameters

**config** nint

The config to modify.

**presencePenalty** float

A scalar subtracted from a token's logit if that token has appeared at least once inside the generated window history. Positive values discourage repetition, while negative values reward repeating tokens (OpenAI style). Defaults to 0.0f.

## litert\_lm\_repetition\_penalty\_config\_set\_repetition\_penalty(nint, float)

Sets the multiplicative repetition penalty for the repetition penalty config.

```
public static extern void  
litert_lm_repetition_penalty_config_set_repetition_penalty(nint config,  
float repetitionPenalty)
```

## Parameters

**config** nint

The config to modify.

**repetitionPenalty** float

A multiplicative penalty applied to a token's logit if that token has appeared at least once inside the generated window history (e.g., 1.0 = no penalty, 1.2 = moderate penalty). Positive logits are divided by this parameter, and negative logits are multiplied (HuggingFace style). The parameter must be  $\geq 1.0f$ ; values less than 1.0f are automatically clamped to 1.0f during execution.

## litert\_lm\_repetition\_penalty\_config\_set\_window\_size(nint, int)

Sets the window size for the repetition penalty config.

```
public static extern void litert_lm_repetition_penalty_config_set_window_size(nint  
config, int windowSize)
```

## Parameters

**config** nint

The config to modify.

**windowSize** int

The maximum number of recent tokens in generation history to consider when computing penalization. Tokens generated prior to this window are forgotten. A value of 0 means tracking all infinite generation history. Must be  $\geq 0$ ; negative values are clamped to 0 during execution.

# Class NativeAPI.Responses

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.Responses
```

## Inheritance

object ← NativeAPI.Responses

## Methods

### litert\_lm\_responses\_delete(nint)

Destroys a LiteRT LM Responses object.

```
public static extern void litert_lm_responses_delete(nint responses)
```

## Parameters

**responses** nint

The responses to destroy.

### litert\_lm\_responses\_get\_num\_candidates(nint)

Returns the number of response candidates.

```
public static extern int litert_lm_responses_get_num_candidates(nint responses)
```

## Parameters

**responses** nint

The responses object.

## Returns

int

The number of candidates.

## litert\_lm\_responses\_get\_num\_token\_scores\_at(nint, int)

Returns the number of tokens for which scores are present at a given index.

```
public static extern int litert_lm_responses_get_num_token_scores_at(nint responses,
int index)
```

### Parameters

**responses** nint

The responses object.

**index** int

The index of the response.

### Returns

int

The number of token scores. Returns 0 if index is out of bounds or no token scores are present.

## litert\_lm\_responses\_get\_response\_text\_at(nint, int)

Returns the response text at a given index.

```
public static extern nint litert_lm_responses_get_response_text_at(nint responses,
int index)
```

### Parameters

**responses** nint

The responses object.

**index** int

The index of the response.

## Returns

nint

The response text. The returned string is owned by the Responses object and is valid only for its lifetime. Returns nint.Zero if index is out of bounds.

## litert\_lm\_responses\_get\_score\_at(nint, int)

Returns the score at a given index.

```
public static extern float litert_lm_responses_get_score_at(nint responses,
int index)
```

## Parameters

**responses** nint

The responses object.

**index** int

The index of the response.

## Returns

float

The score. Returns 0.0f if index is out of bounds or no score is present.

## litert\_lm\_responses\_get\_token\_length\_at(nint, int)

Returns the token length at a given index.

```
public static extern int litert_lm_responses_get_token_length_at(nint responses,
int index)
```

## Parameters

**responses** nint

The responses object.

**index** int

The index of the response.

## Returns

int

The token length. Returns 0 if index is out of bounds or no token length is present.

## litert\_lm\_responses\_get\_token\_scores\_at(nint, int)

Returns the token scores at a given index.

```
public static extern nint litert_lm_responses_get_token_scores_at(nint responses,
int index)
```

## Parameters

**responses** nint

The responses object.

**index** int

The index of the response.

## Returns

nint

A pointer to the internal array of token scores. Returns nint.Zero if index is out of bounds or no token scores are present.

## litert\_lm\_responses\_has\_score\_at(nint, int)

Returns whether the response contains a score at the given index.

```
public static extern bool litert_lm_responses_has_score_at(nint responses,
int index)
```

## Parameters

**responses** nint

The responses object.

**index** int

The index of the response.

## Returns

bool

[true](#) if the score is available at the given index, [false](#) otherwise.

## litert\_lm\_responses\_has\_token\_length\_at(nint, int)

Returns whether the response contains a token length at the given index.

```
public static extern bool litert_lm_responses_has_token_length_at(nint responses,
int index)
```

## Parameters

**responses** nint

The responses object.

**index** int

The index of the response.

## Returns

bool



[true](#) if the token length is available at the given index, [false](#) otherwise.

## litert\_lm\_responses\_has\_token\_scores\_at(nint, int)

Returns whether the response contains token scores at the given index.

```
public static extern bool litert_lm_responses_has_token_scores_at(nint responses,
int index)
```

### Parameters

**responses** nint

The responses object.

**index** int

The index of the response.

### Returns

bool

[true](#) if token scores are available at the given index, [false](#) otherwise.

# Class NativeAPI.SamplerParams

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.SamplerParams
```

## Inheritance

object ← NativeAPI.SamplerParams

## Methods

### litert\_lm\_sampler\_params\_create(SamplerType)

Creates LiteRT LM Sampler Parameters with a specific sampler type. The caller is responsible for destroying the parameters using [litert\\_lm\\_sampler\\_params\\_delete\(nint\)](#).

```
public static extern nint litert_lm_sampler_params_create(SamplerType type)
```

## Parameters

type [SamplerType](#)

The sampler type to use.

## Returns

nint

A pointer to the created parameters, or nint.Zero on failure.

### litert\_lm\_sampler\_params\_delete(nint)

Destroys LiteRT LM Sampler Parameters.

```
public static extern void litert_lm_sampler_params_delete(nint @params)
```

## Parameters

**params** nint

The parameters to destroy.

## litert\_lm\_sampler\_params\_set\_seed(nint, int)

Sets the seed.

```
public static extern void litert_lm_sampler_params_set_seed(nint @params, int seed)
```

## Parameters

**params** nint

**seed** int

## litert\_lm\_sampler\_params\_set\_temperature(nint, float)

Sets the temperature.

```
public static extern void litert_lm_sampler_params_set_temperature(nint @params, float temperature)
```

## Parameters

**params** nint

**temperature** float

## litert\_lm\_sampler\_params\_set\_top\_k(nint, int)

Sets the top-k value.

```
public static extern void litert_lm_sampler_params_set_top_k(nint @params, int topK)
```

## Parameters

**params** nint

**topK** int

## litert\_lm\_sampler\_params\_set\_top\_p(nint, float)

Sets the top-p value.

```
public static extern void litert_lm_sampler_params_set_top_p(nint @params,  
float topP)
```

## Parameters

**params** nint

**topP** float

# Class NativeAPI.Session

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.Session
```

## Inheritance

object ← NativeAPI.Session

## Methods

### litert\_lm\_session\_cancel\_process(nint)

Cancels the current processing in the session.

```
public static extern void litert_lm_session_cancel_process(nint session)
```

## Parameters

**session** nint

The session to cancel processing on.

### litert\_lm\_session\_delete(nint)

Destroys a LiteRT LM Session.

```
public static extern void litert_lm_session_delete(nint session)
```

## Parameters

**session** nint

The session to destroy.

# litert\_lm\_session\_generate\_content(nint, nint[], nuint)

Generates content from the input prompt.

```
public static extern nint litert_lm_session_generate_content(nint session, nint[]  
inputs, nuint numInputs)
```

## Parameters

**session** nint

The session to use for generation.

**inputs** nint[]

An array of InputData structs representing the multimodal input.

**numInputs** nuint

The number of InputData structs in the array.

## Returns

nint

A pointer to the responses, or nint.Zero on failure. The caller is responsible for deleting the responses using [litert\\_lm\\_responses\\_delete\(nint\)](#).

# litert\_lm\_session\_generate\_content\_stream(nint, nint[], nuint, nint, nint)

Generates content from the input prompt and streams the response via a callback. This is a non-blocking call that will invoke the callback from a background thread for each chunk.

```
public static extern int litert_lm_session_generate_content_stream(nint session,  
nint[] inputs, nuint numInputs, nint callback, nint callbackData)
```

## Parameters

**session** nint

The session to use for generation.

**inputs** nint[]

An array of InputData structs representing the multimodal input.

**numInputs** nuint

The number of InputData structs in the array.

**callback** nint

The callback function ([NativeAPI.StreamCallback](#)) to receive response chunks.

**callbackData** nint

A pointer to user data that will be passed to the callback.

## Returns

int

0 on success, non-zero on failure to start the stream.

## litert\_lm\_session\_get\_benchmark\_info(nint)

Retrieves the benchmark information from the session. The caller is responsible for destroying the benchmark info using [litert\\_lm\\_benchmark\\_info\\_delete\(nint\)](#).

```
public static extern nint litert_lm_session_get_benchmark_info(nint session)
```

## Parameters

**session** nint

The session to get the benchmark info from.

## Returns

nint

A pointer to the benchmark info, or nint.Zero on failure.

## litert\_lm\_session\_rewind\_to\_checkpoint(nint, string)

Rewinds the session to the given checkpoint label.

```
public static extern int litert_lm_session_rewind_to_checkpoint(nint session,
string label)
```

### Parameters

**session** nint

The session to rewind.

**label** string

Label of the checkpoint to rewind to.

### Returns

int

0 on success, non-zero on failure.

## litert\_lm\_session\_rewind\_to\_step(nint, int)

Rewinds the session to a specific step number.

```
public static extern int litert_lm_session_rewind_to_step(nint session, int step)
```

### Parameters

**session** nint

The session to rewind.

**step** int

The step number to rewind to.

### Returns



int

0 on success, non-zero on failure.

## litert\_lm\_session\_run\_decode(nint)

Starts the decoding process for the model to predict the response based on the input prompt/query added after using [litert\\_lm\\_session\\_run\\_prefill\(nint, nint\[\], nuint\)](#). This is a blocking call and the function will return when the decoding process is done.

```
public static extern nint litert_lm_session_run_decode(nint session)
```

### Parameters

**session** nint

The session to use.

### Returns

nint

A pointer to the responses, or nint.Zero on failure. The caller is responsible for deleting the responses using [litert\\_lm\\_responses\\_delete\(nint\)](#).

## litert\_lm\_session\_run\_decode\_async(nint, nint, nint)

Starts the decoding process for the model to predict the response based on the input prompt/query added after using [litert\\_lm\\_session\\_run\\_prefill\(nint, nint\[\], nuint\)](#). This is a non-blocking call that will stream responses via a callback.

```
public static extern int litert_lm_session_run_decode_async(nint session, nint  
callback, nint callbackData)
```

### Parameters

**session** nint

The session to use.

**callback** nint

The callback function ([NativeAPI.StreamCallback](#)) to receive response chunks.

**callbackData** nint

A pointer to user data that will be passed to the callback.

## Returns

int

0 on success, non-zero on failure.

## litert\_lm\_session\_run\_prefill(nint, nint[], nuint)

Adds the input prompt/query to the model for starting the prefilling process. This is a blocking call and the function will return when the prefill process is done.

```
public static extern int litert_lm_session_run_prefill(nint session, nint[] inputs,
nuint numInputs)
```

## Parameters

**session** nint

The session to use.

**inputs** nint[]

An array of InputData structs representing the multimodal input.

**numInputs** nuint

The number of InputData structs in the array.

## Returns

int

0 on success, non-zero on failure.

## litert\_lm\_session\_run\_text\_scoring(nint, string[], nuint, bool)

Scores the target text after the prefill process is done.

```
public static extern nint litert_lm_session_run_text_scoring(nint session, string[]
targetText, nuint numTargets, bool storeTokenLengths)
```

### Parameters

**session** nint

The session to use.

**targetText** string[]

An array of target text strings to score.

**numTargets** nuint

The number of strings in the targetText array.

**storeTokenLengths** bool

Whether to store the token lengths of the target texts in the responses.

### Returns

nint

A pointer to the responses, or nint.Zero on failure. The caller is responsible for deleting the responses using [litert\\_lm\\_responses\\_delete\(nint\)](#).

## litert\_lm\_session\_save\_checkpoint(nint, string)

Saves the current state of the session to a checkpoint with the given label.

```
public static extern int litert_lm_session_save_checkpoint(nint session,
string label)
```

### Parameters

`session` nint

The session to save checkpoint for.

`label` string

Label for the checkpoint.

Returns

int

0 on success, non-zero on failure.

# Class NativeAPI.SessionConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.SessionConfig
```

## Inheritance

object ← NativeAPI.SessionConfig

## Methods

### litert\_lm\_session\_config\_create()

Creates a LiteRT LM Session Config. The caller is responsible for destroying the config using [litert\\_lm\\_session\\_config\\_delete\(nint\)](#).

```
public static extern nint litert_lm_session_config_create()
```

## Returns

nint

A pointer to the created config, or nint.Zero on failure.

### litert\_lm\_session\_config\_delete(nint)

Destroys a LiteRT LM Session Config.

```
public static extern void litert_lm_session_config_delete(nint config)
```

## Parameters

**config** nint

The config to destroy.

## litert\_lm\_session\_config\_set\_apply\_prompt\_template(nint, bool)

Sets whether to apply prompt template for this session.

```
public static extern void litert_lm_session_config_set_apply_prompt_template(nint config, bool applyPromptTemplate)
```

### Parameters

**config** nint

The config to modify.

**applyPromptTemplate** bool

Whether to apply prompt template.

## litert\_lm\_session\_config\_set\_audio\_lora\_path(nint, string)

Sets the path to the Audio LoRA weights file.

```
public static extern int litert_lm_session_config_set_audio_lora_path(nint config, string audioLoraPath)
```

### Parameters

**config** nint

The config to modify.

**audioLoraPath** string

The path to the audio LoRA weights file.

### Returns

int

0 on success, non-zero on failure.

# litert\_lm\_session\_config\_set\_lora\_path(nint, string)

Sets the path to the LoRA weights file.

```
public static extern int litert_lm_session_config_set_lora_path(nint config,
string loraPath)
```

## Parameters

**config** nint

The config to modify.

**loraPath** string

The path to the text LoRA weights file.

## Returns

int

0 on success, non-zero on failure.

# litert\_lm\_session\_config\_set\_max\_output\_tokens(nint, int)

Sets the maximum number of output tokens per decode step for this session.

```
public static extern void litert_lm_session_config_set_max_output_tokens(nint
config, int maxOutputTokens)
```

## Parameters

**config** nint

The config to modify.

**maxOutputTokens** int

The maximum number of output tokens.

# litert\_lm\_session\_config\_set\_sampler\_params(nint, nint)

Sets the sampler parameters for this session config.

```
public static extern void litert_lm_session_config_set_sampler_params(nint config,  
nint samplerParams)
```

## Parameters

**config** nint

The config to modify.

**samplerParams** nint

The sampler parameters to use.



# Delegate NativeAPI.StreamCallback

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

Callback for streaming responses.

```
public delegate void NativeAPI.StreamCallback(nint callbackData, nint chunk)
```

## Parameters

**callbackData** nint

A pointer to user-defined data passed to the stream function.

**chunk** nint

A pointer to the stream chunk object. It's only valid for the duration of the call.

# Class NativeAPI.StreamChunk

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.StreamChunk
```

## Inheritance

object ← NativeAPI.StreamChunk

## Methods

### litert\_lm\_stream\_chunk\_get\_error(nint)

Gets the error message associated with this chunk, if any.

```
public static extern nint litert_lm_stream_chunk_get_error(nint chunk)
```

## Parameters

**chunk** nint

## Returns

nint

## Remarks

Returns nint.Zero if there is no error.

### litert\_lm\_stream\_chunk\_get\_text(nint)

Gets the text content of the chunk.

```
public static extern nint litert_lm_stream_chunk_get_text(nint chunk)
```

## Parameters

**chunk** nint

Returns

nint

Remarks

The returned string is owned by the chunk and is only valid as long as the chunk is valid. Returns nint.Zero if there is no text content in this chunk (e.g. if it is an error or metadata-only chunk).

## litert\_lm\_stream\_chunk\_is\_final(nint)

Returns [true](#) if this is the final chunk of the stream.

```
public static extern bool litert_lm_stream_chunk_is_final(nint chunk)
```

Parameters

**chunk** nint

Returns

bool

# Class NativeAPI.SuppressTokensConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.SuppressTokensConfig
```

## Inheritance

object ← NativeAPI.SuppressTokensConfig

## Methods

### litert\_lm\_suppress\_tokens\_config\_create()

Creates a LiteRT LM Suppress Tokens Config with default values (an empty set of suppressed tokens, which means token suppression is disabled).

```
public static extern nint litert_lm_suppress_tokens_config_create()
```

## Returns

nint

A pointer to the created config, or IntPtr.Zero on failure.

## Remarks

When `suppress_tokens` is configured with one or more token IDs, the logits corresponding to those exact token IDs will be set directly to -inf during generation. This guarantees that those tokens can never be sampled by the model.

The caller is responsible for destroying the config using [litert\\_lm\\_suppress\\_tokens\\_config\\_delete\(nint\)](#).

### litert\_lm\_suppress\_tokens\_config\_delete(nint)

Destroys a LiteRT LM Suppress Tokens Config.

```
public static extern void litert_lm_suppress_tokens_config_delete(nint config)
```

## Parameters

**config** nint

The config to destroy.

## litert\_lm\_suppress\_tokens\_config\_set\_suppress\_tokens(nint, int[]?, nuint)

Sets the list of token IDs to suppress for the suppress tokens config.

```
public static extern void litert_lm_suppress_tokens_config_set_suppress_tokens(nint config, int[]? suppressTokens, nuint numTokens)
```

## Parameters

**config** nint

The config to modify.

**suppressTokens** int[]

An array of integer token IDs that should be banned from generation. During every decode step, each listed token ID's candidate logit will be forced to -inf. If **suppress\_tokens** is [null](#) or **num\_tokens** is 0, any previously set suppressed tokens are cleared and token suppression is disabled.

**numTokens** nuint

The number of token IDs in the **suppress\_tokens** array.

# Class NativeAPI.ThinkingConfig

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.ThinkingConfig
```

## Inheritance

object ← NativeAPI.ThinkingConfig

## Methods

### litert\_lm\_thinking\_config\_create()

Creates a default LiteRT LM Thinking Config (enabled with infinite budget -1). The caller is responsible for destroying the config using [litert\\_lm\\_thinking\\_config\\_delete\(nint\)](#).

```
public static extern nint litert_lm_thinking_config_create()
```

## Returns

nint

A pointer to the created config, or nint.Zero on failure.

### litert\_lm\_thinking\_config\_delete(nint)

Destroys a LiteRT LM Thinking Config.

```
public static extern void litert_lm_thinking_config_delete(nint config)
```

## Parameters

**config** nint

The config to destroy.

## litert\_lm\_thinking\_config\_set\_enable\_thinking(nint, bool)

Sets whether thinking/reasoning generation is enabled.

```
public static extern void litert_lm_thinking_config_set_enable_thinking(nint config,
bool enableThinking)
```

### Parameters

**config** nint

The config to modify.

**enableThinking** bool

Whether thinking is enabled.

## litert\_lm\_thinking\_config\_set\_thinking\_token\_budget(nint, int)

Sets the thinking token budget.

```
public static extern void litert_lm_thinking_config_set_thinking_token_budget(nint
config, int thinkingTokenBudget)
```

### Parameters

**config** nint

The config to modify.

**thinkingTokenBudget** int

Budget for token-by-token reasoning generation (-1 for infinite).

# Class NativeAPI.TokenUnion

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.TokenUnion
```

## Inheritance

object ← NativeAPI.TokenUnion

## Methods

### litert\_lm\_token\_union\_delete(nint)

Destroys a LiteRT LM Token Union.

```
public static extern void litert_lm_token_union_delete(nint tokenUnion)
```

## Parameters

**tokenUnion** nint

The token union to destroy.

### litert\_lm\_token\_union\_get\_ids(nint, out nint, out nuint)

Returns the token ids from a token union.

```
public static extern int litert_lm_token_union_get_ids(nint tokenUnion, out nint  
outTokens, out nuint outNumTokens)
```

## Parameters

**tokenUnion** nint

The token union.

**outTokens** nint



A pointer to receive the internal array of token ids. The received pointer is valid only for the lifetime of the TokenUnion object.

`outNumTokens` `nuint`

A pointer to receive the number of token ids.

## Returns

`int`

0 on success, non-zero if the type is not [Ids](#).

## `litert_lm_token_union_get_string(nint)`

Returns the string value from a token union.

```
public static extern nint litert_lm_token_union_get_string(nint tokenUnion)
```

## Parameters

`tokenUnion` `nint`

The token union.

## Returns

`nint`

The string value, or `nint.Zero` if the type is not [String](#). The returned string is owned by the TokenUnion object and is valid only for its lifetime.

## `litert_lm_token_union_get_type(nint)`

Returns the type of the token union.

```
public static extern TokenUnionType litert_lm_token_union_get_type(nint tokenUnion)
```

## Parameters

`tokenUnion` nint

The token union.

Returns

[TokenUnionType](#)

The type of the token union.

# Class NativeAPI.TokenUnions

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.TokenUnions
```

## Inheritance

object ← NativeAPI.TokenUnions

## Methods

### litert\_lm\_token\_unions\_delete(nint)

Destroys a LiteRT LM Token Unions object.

```
public static extern void litert_lm_token_unions_delete(nint tokens)
```

## Parameters

**tokens** nint

The token unions object to destroy.

### litert\_lm\_token\_unions\_get\_num\_tokens(nint)

Returns the number of token unions in the collection.

```
public static extern nuint litert_lm_token_unions_get_num_tokens(nint tokens)
```

## Parameters

**tokens** nint

The token unions object.

## Returns

nuint

The number of token unions.

## litert\_lm\_token\_unions\_get\_token\_at(nint, nuint)

Returns the token union at a given index from a collection.

```
public static extern nint litert_lm_token_unions_get_token_at(nint tokens,  
nuint index)
```

### Parameters

**tokens** nint

The token unions collection.

**index** nuint

The index of the token union.

### Returns

nint

A pointer to the token union at the given index, or nint.Zero if the index is out of bounds.  
The caller is responsible for deleting the result using [litert\\_lm\\_token\\_union\\_delete\(nint\)](#).

# Class NativeAPI.TokenizeResult

Namespace: [Uralstech.UAI.LiteRTLM.Native](#)

```
public static class NativeAPI.TokenizeResult
```

## Inheritance

object ← NativeAPI.TokenizeResult

## Methods

### litert\_lm\_tokenize\_result\_delete(nint)

Destroys a LiteRT LM Tokenize Result.

```
public static extern void litert_lm_tokenize_result_delete(nint result)
```

## Parameters

**result** nint

The tokenize result to destroy.

### litert\_lm\_tokenize\_result\_get\_num\_tokens(nint)

Returns the number of token ids from a tokenize result.

```
public static extern nuint litert_lm_tokenize_result_get_num_tokens(nint result)
```

## Parameters

**result** nint

The tokenize result.

## Returns

nuint

The number of token ids.

## litert\_lm\_tokenize\_result\_get\_tokens(nint)

Returns the token ids from a tokenize result.

```
public static extern nint litert_lm_tokenize_result_get_tokens(nint result)
```

### Parameters

**result** nint

The tokenize result.

### Returns

nint

A pointer to the internal array of token ids. The returned pointer is valid only for the lifetime of the **result** object.